

BFS Traversal

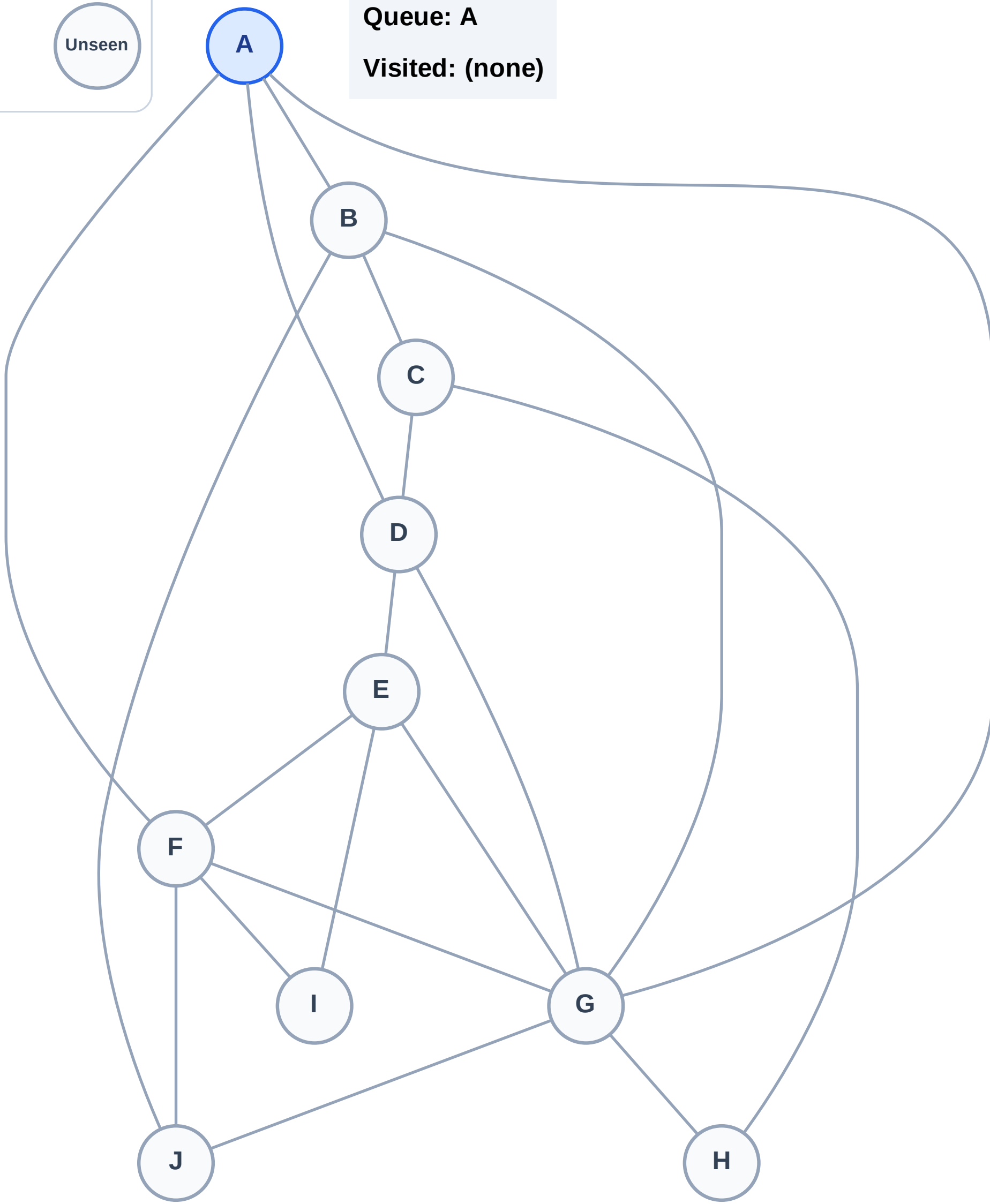
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

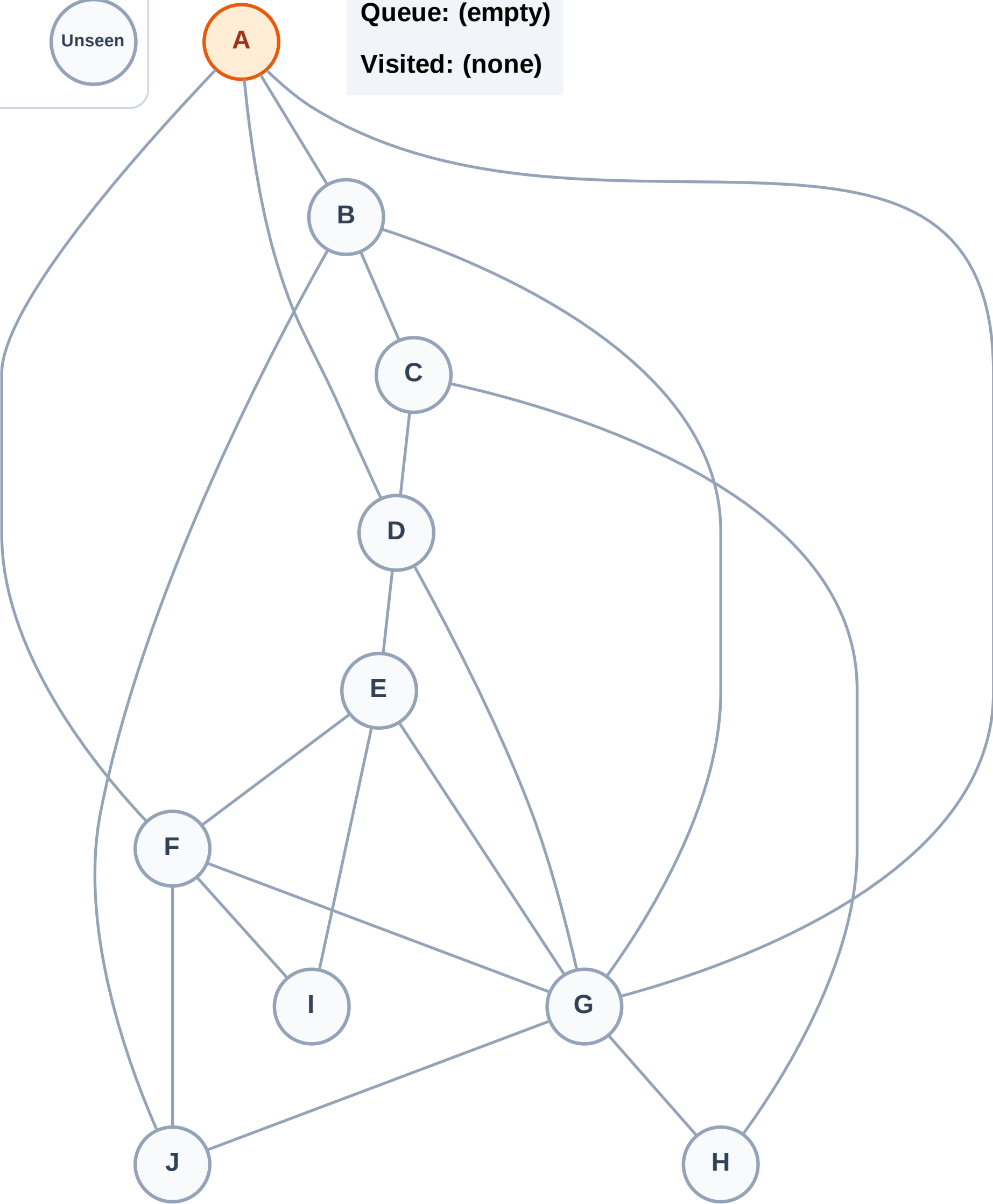
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

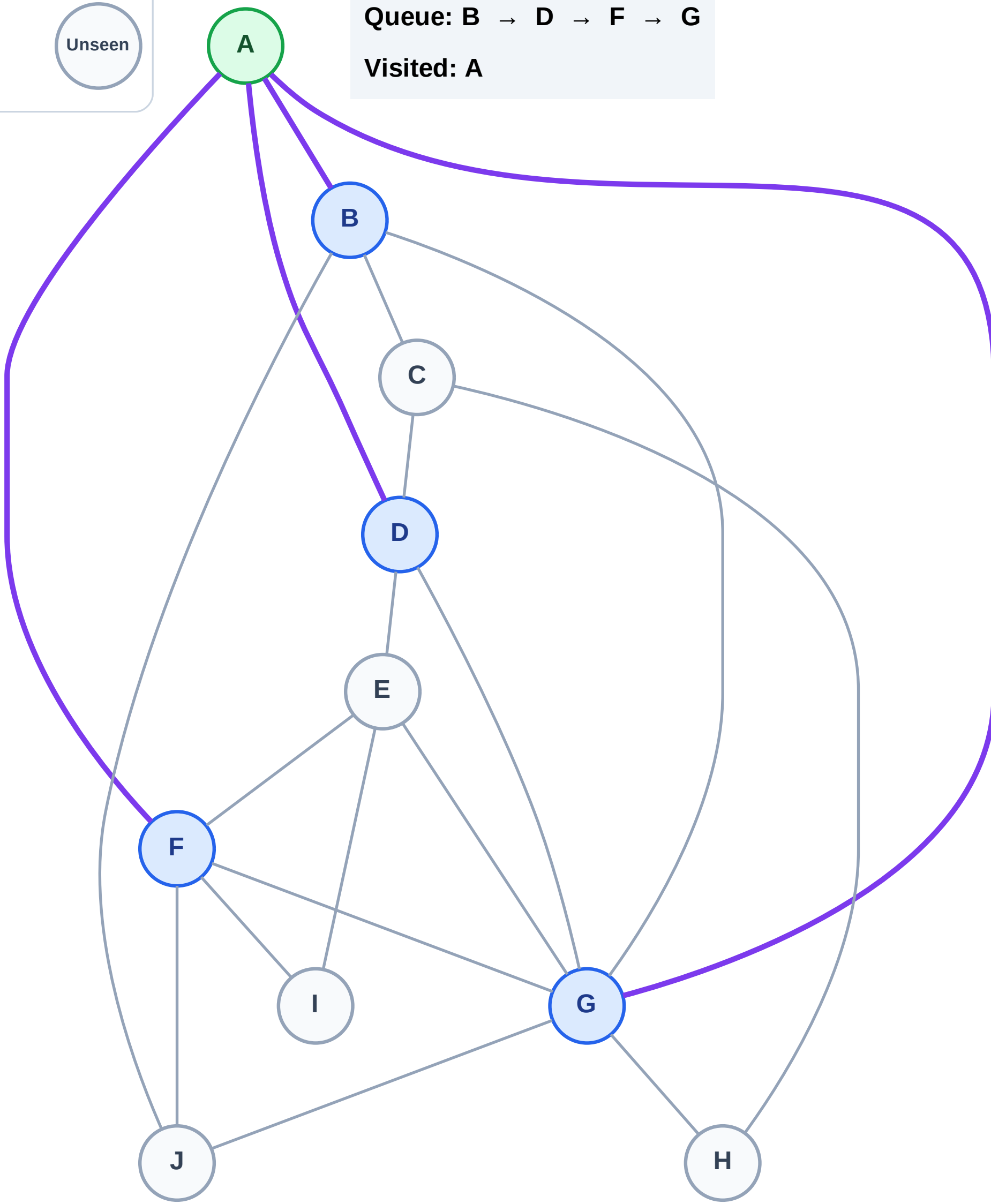
Step 3: Discover B, D, F, G from A

Legend



Queue: B → D → F → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

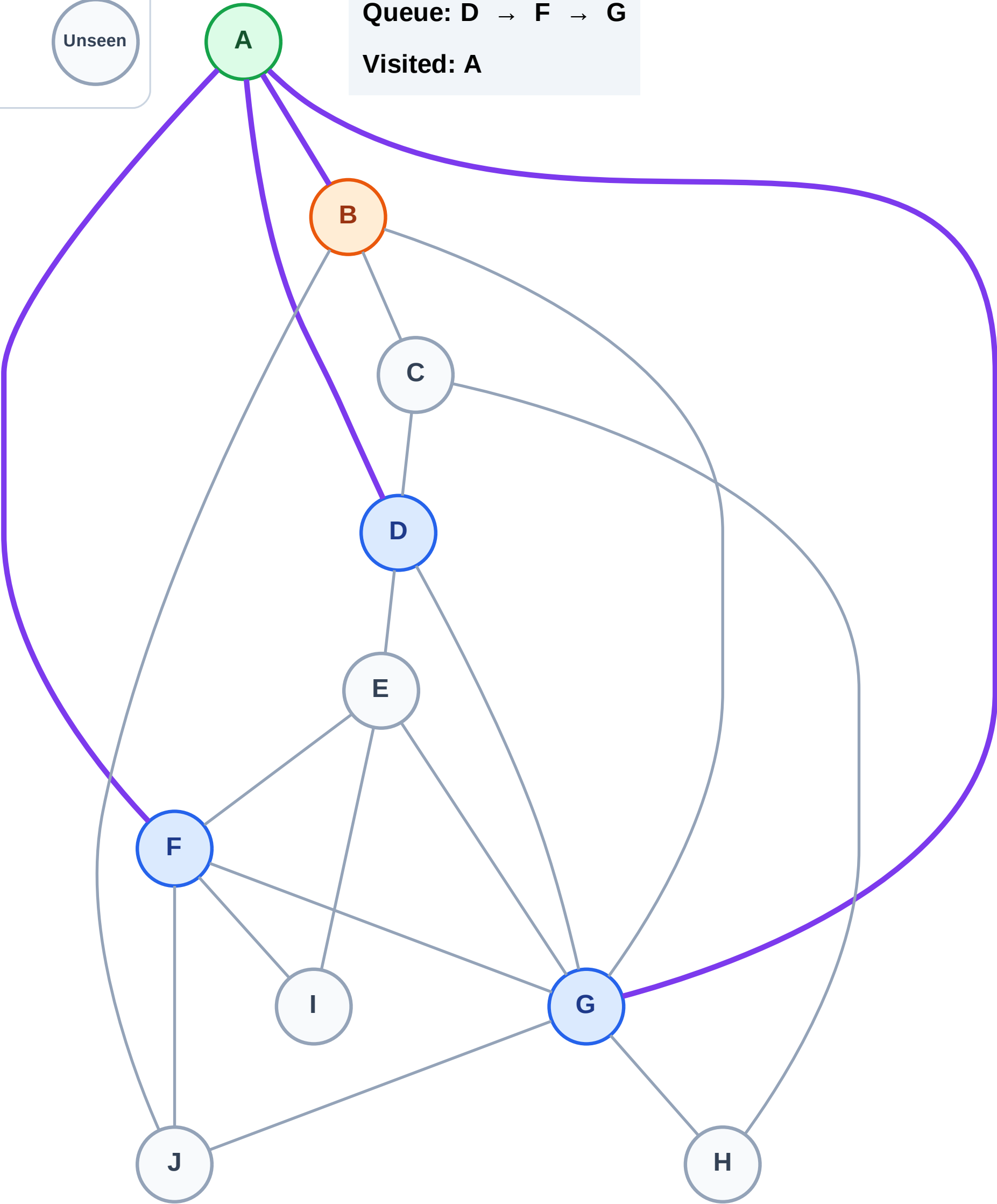
Processing

Discovered

Unseen

Queue: D → F → G

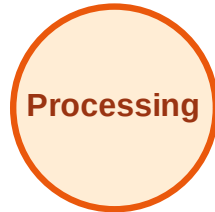
Visited: A



BFS Traversal

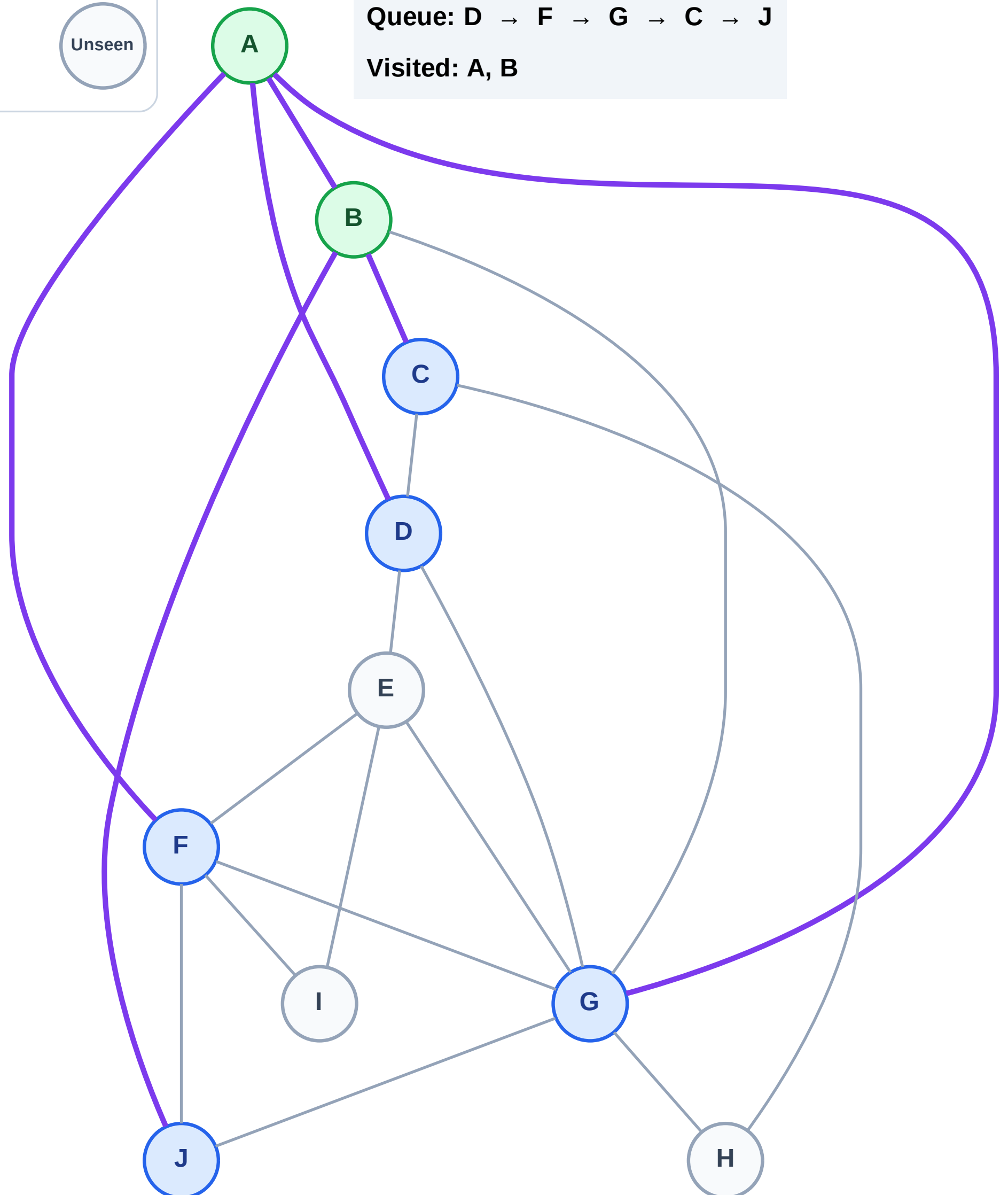
Step 5: Discover C, J from B

Legend



Queue: D → F → G → C → J

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

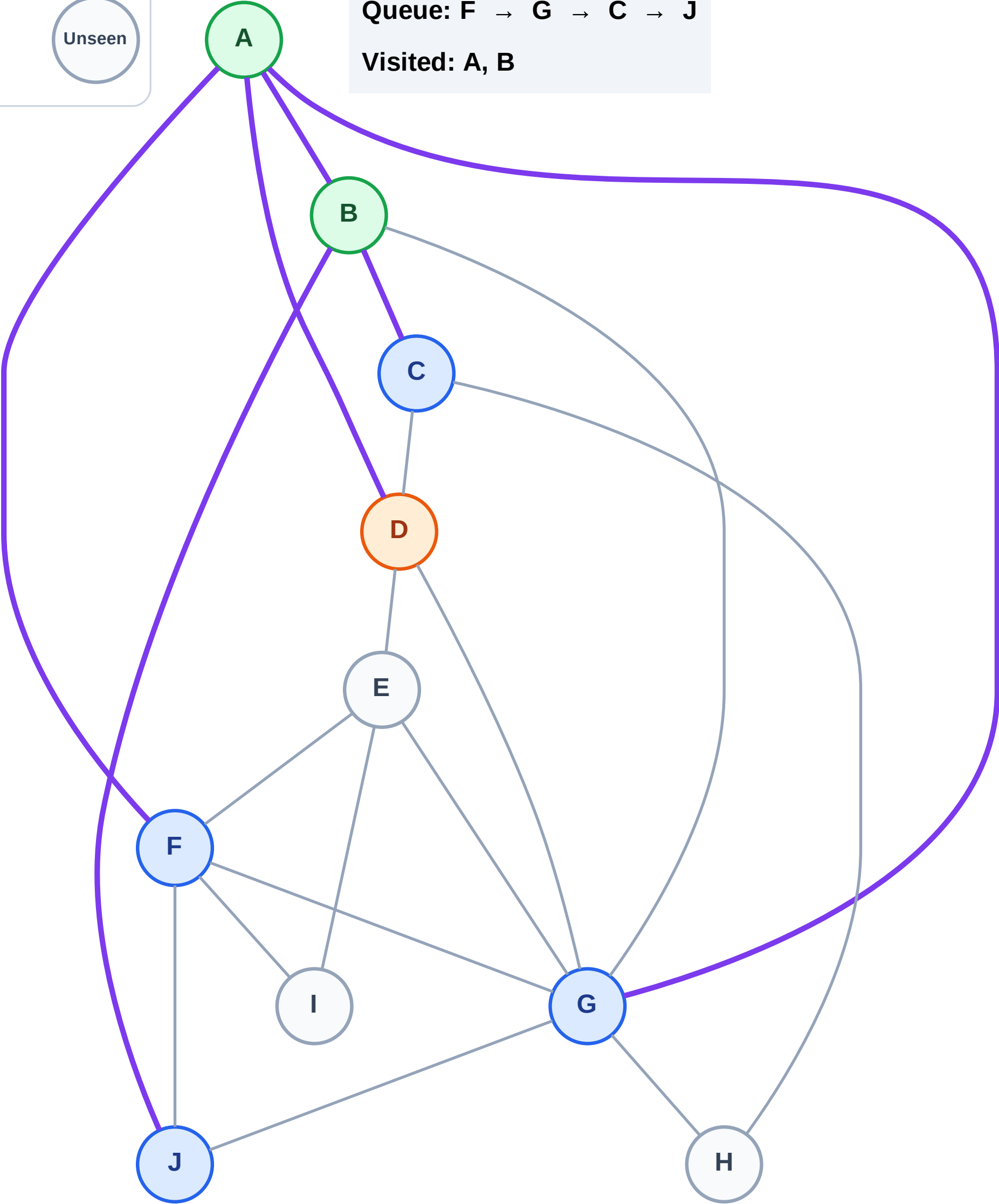
Processing

Discovered

Unseen

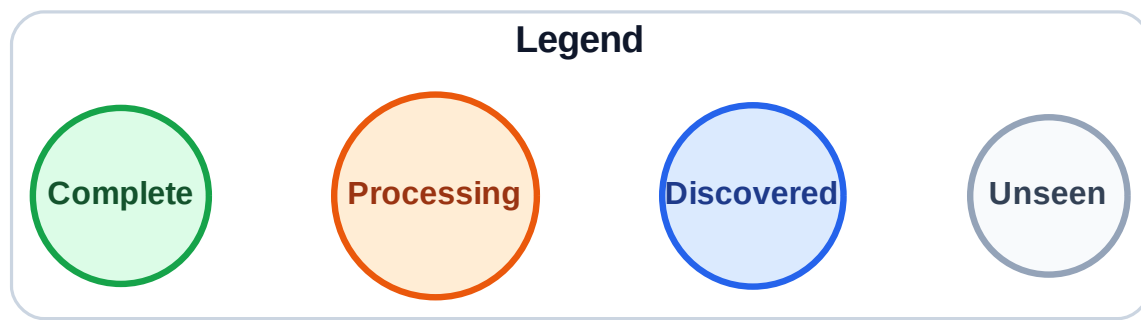
Queue: F → G → C → J

Visited: A, B



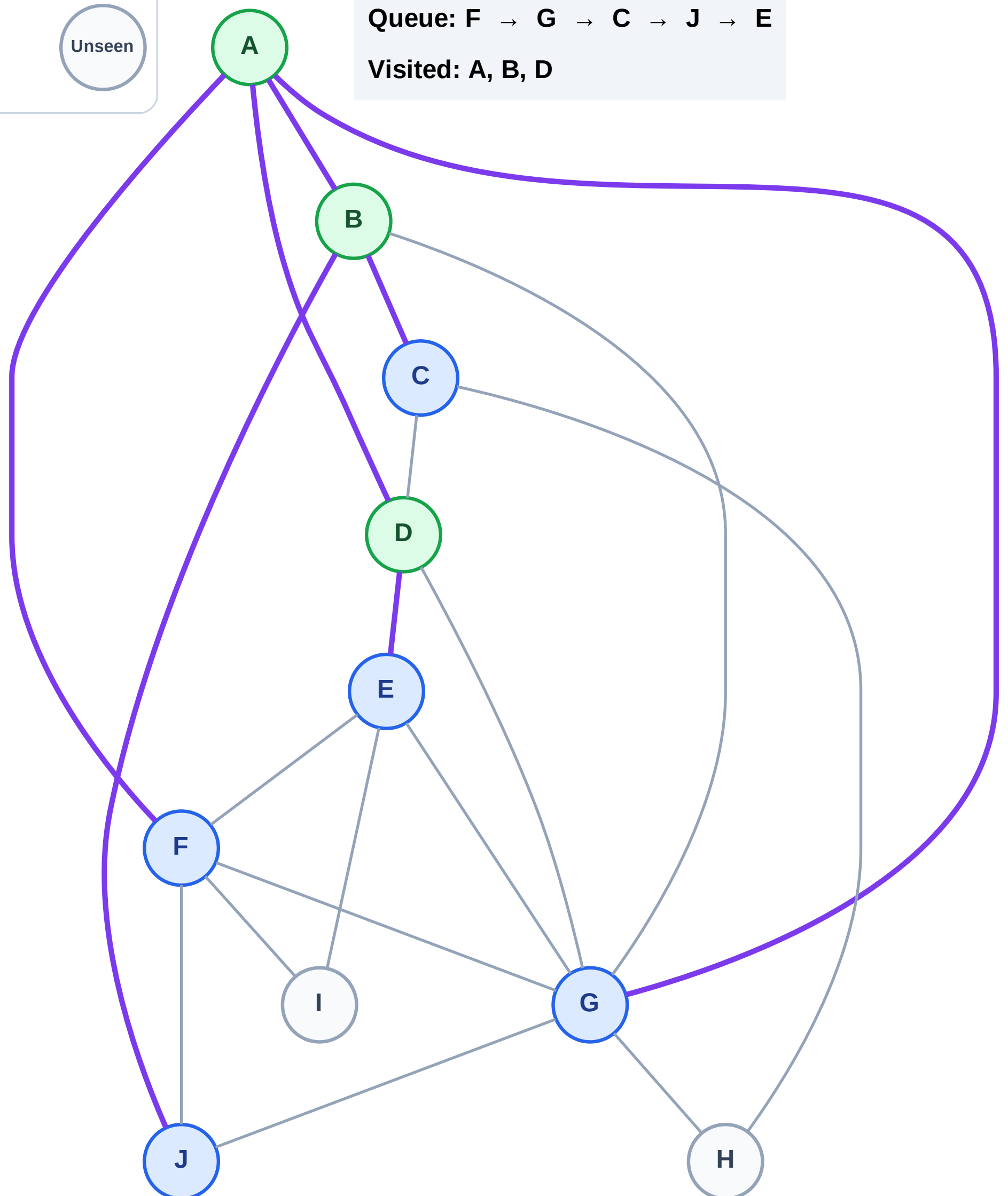
BFS Traversal

Step 7: Discover E from D



Queue: F → G → C → J → E

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

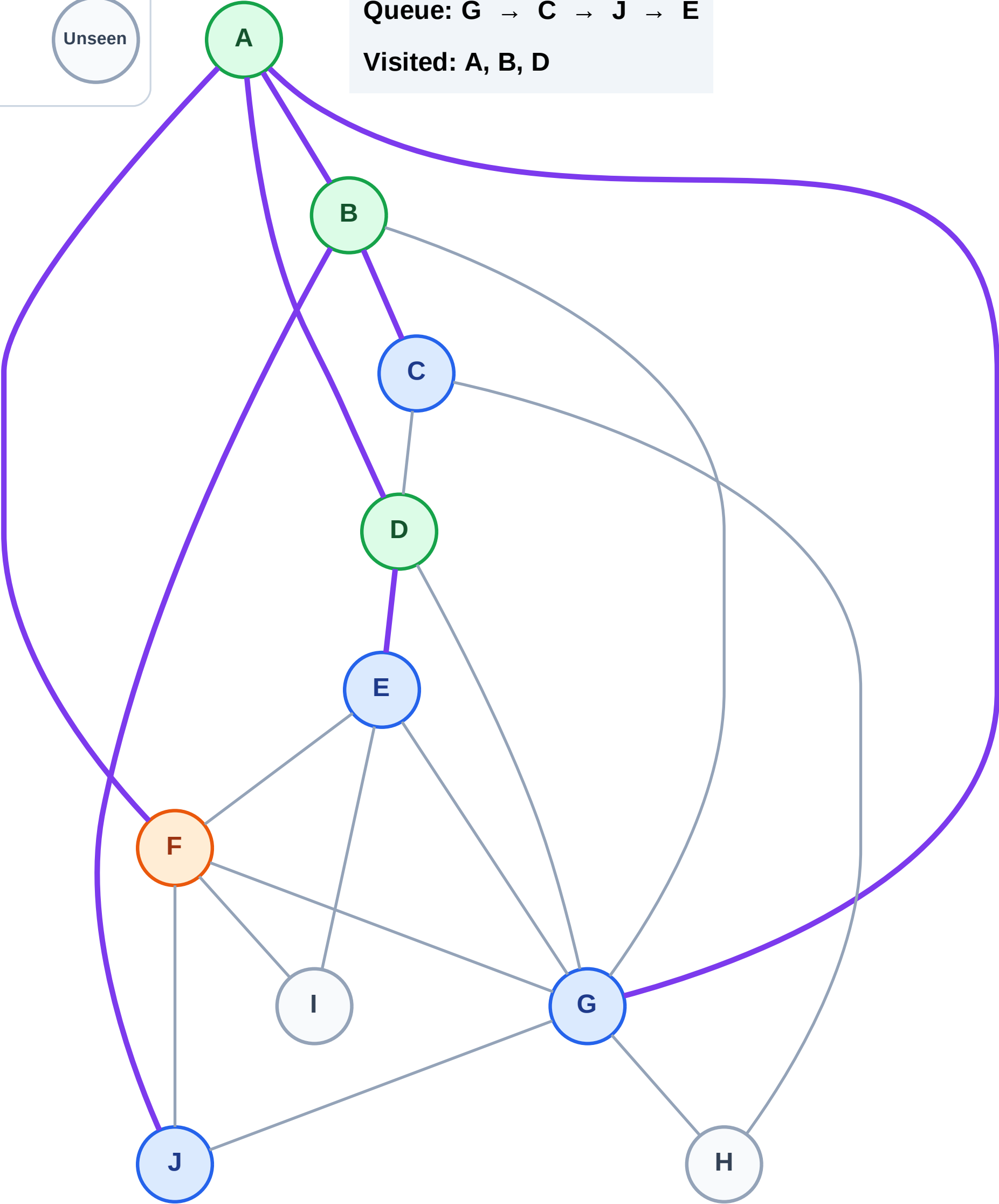
Processing

Discovered

Unseen

Queue: G → C → J → E

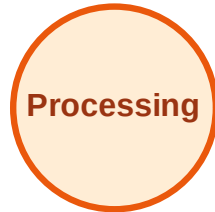
Visited: A, B, D



BFS Traversal

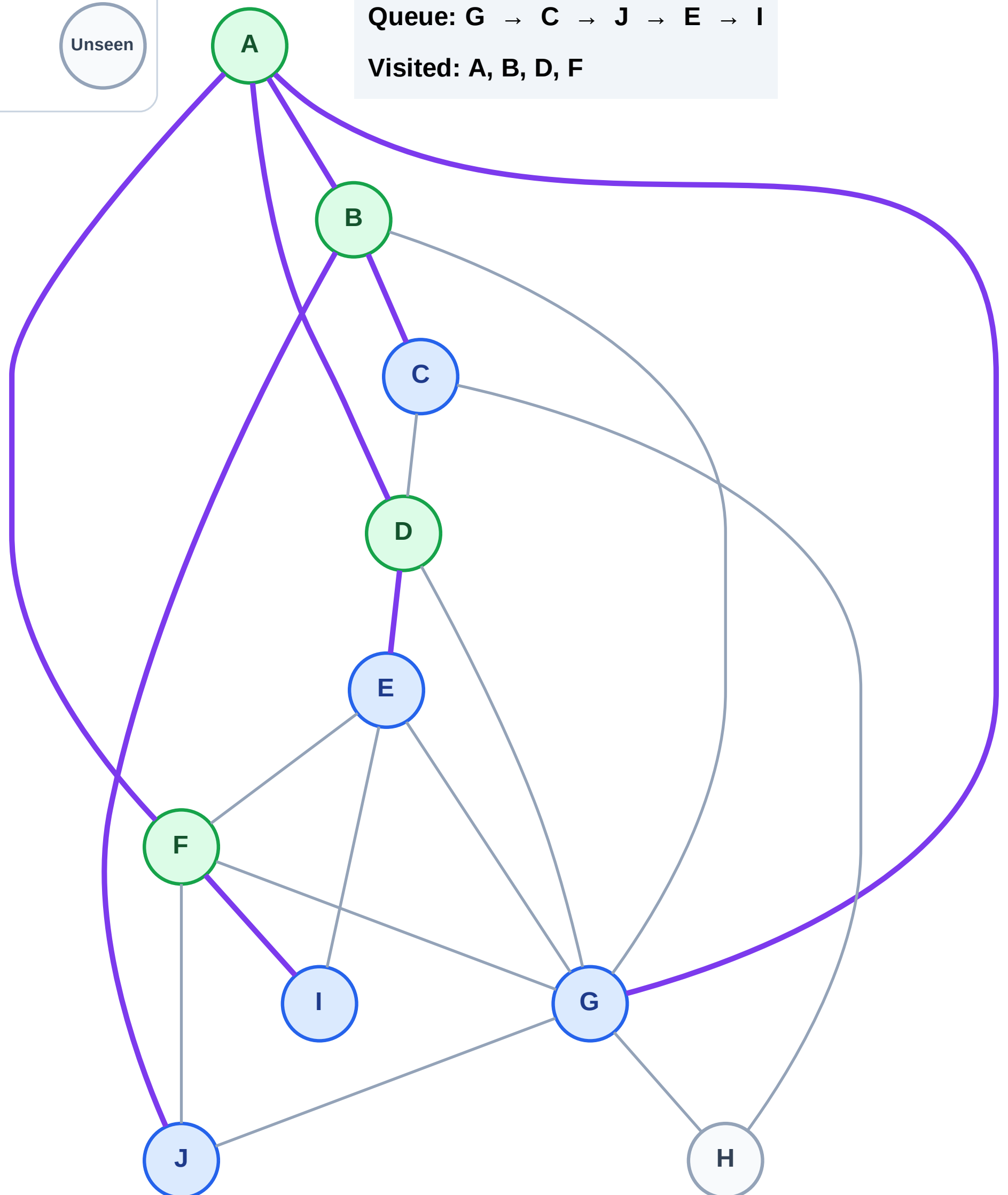
Step 9: Discover I from F

Legend



Queue: G → C → J → E → I

Visited: A, B, D, F



BFS Traversal

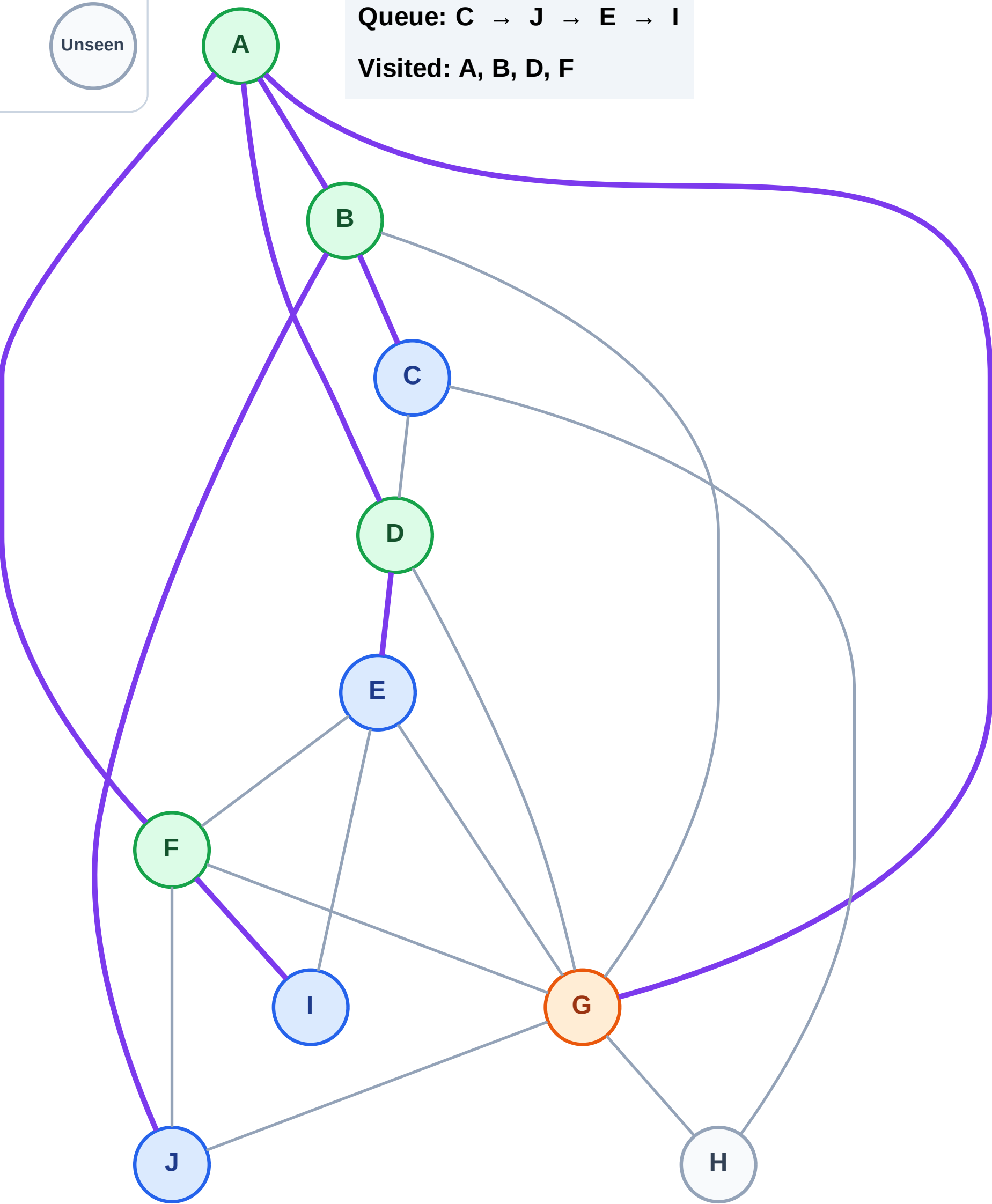
Step 10: Process node G

Legend



Queue: C → J → E → I

Visited: A, B, D, F



BFS Traversal

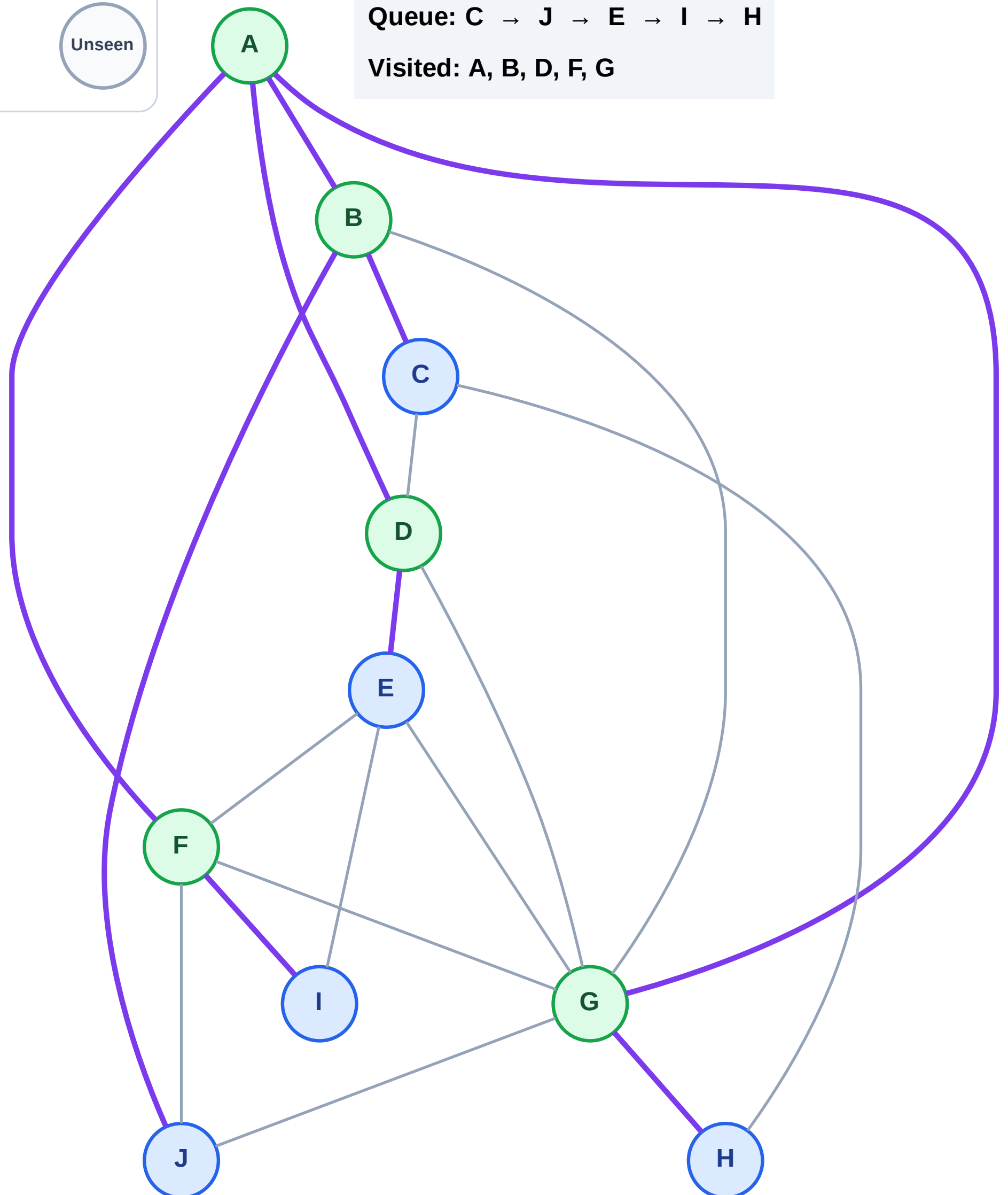
Step 11: Discover H from G

Legend



Queue: C → J → E → I → H

Visited: A, B, D, F, G



BFS Traversal

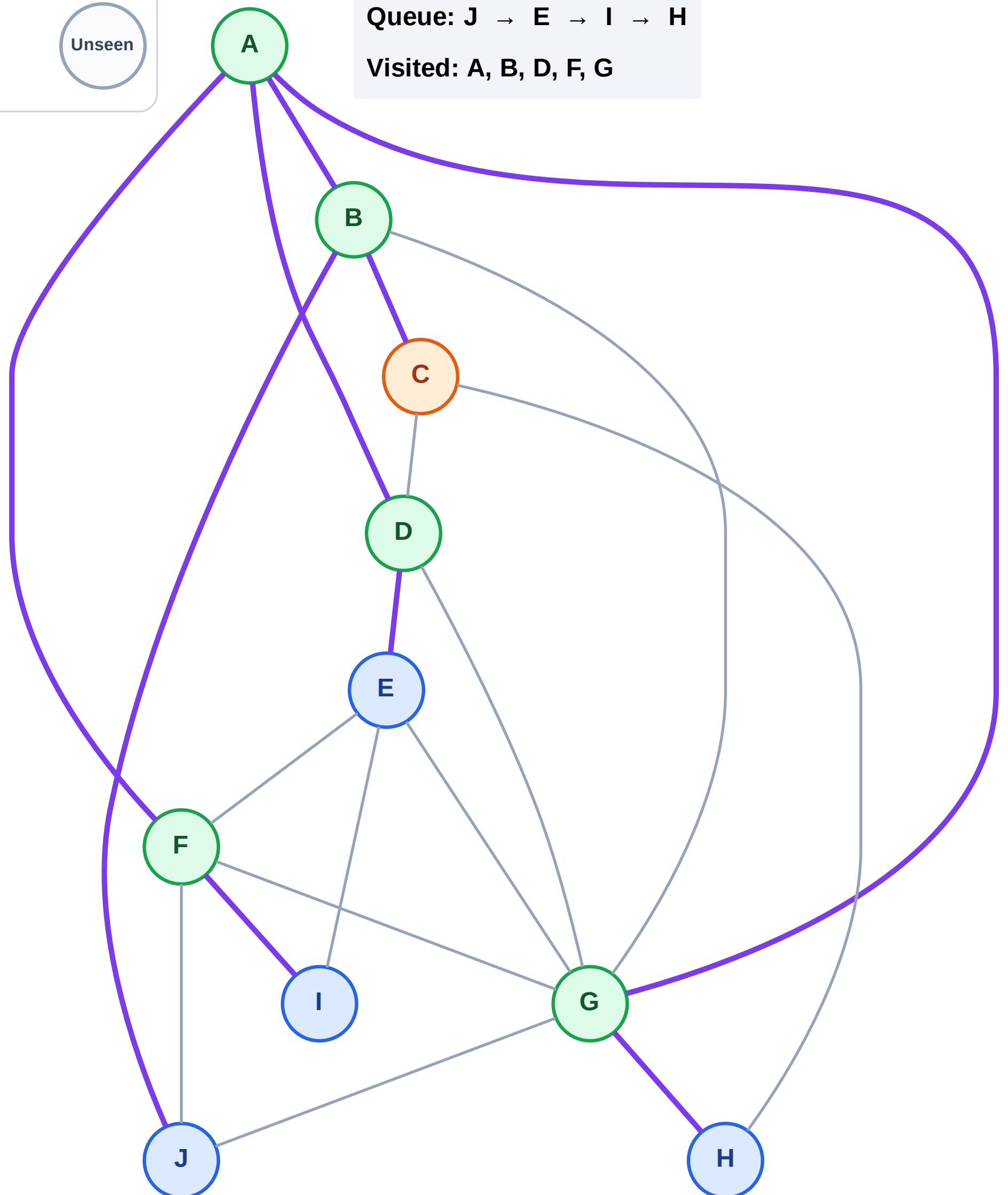
Step 12: Process node C

Legend



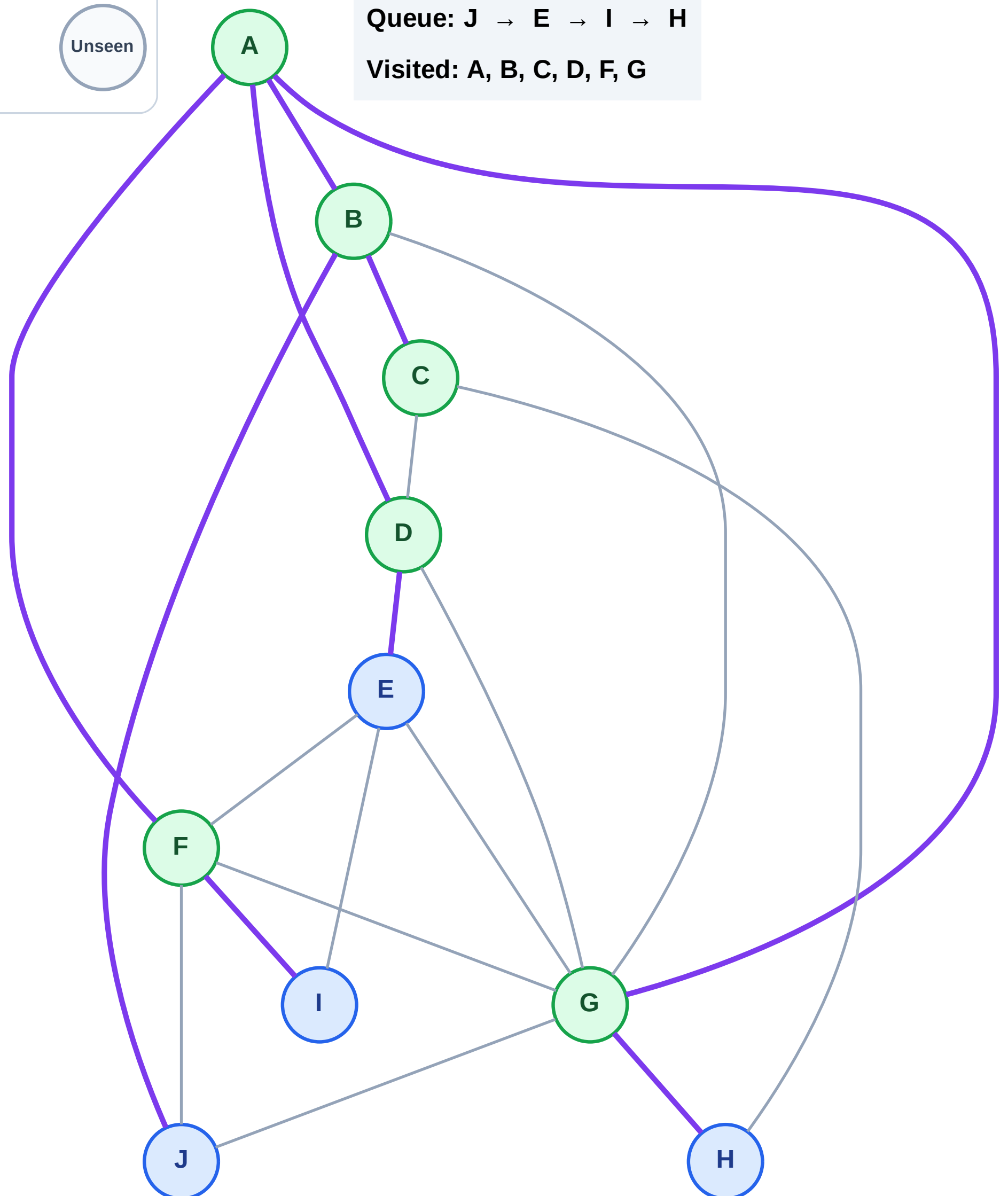
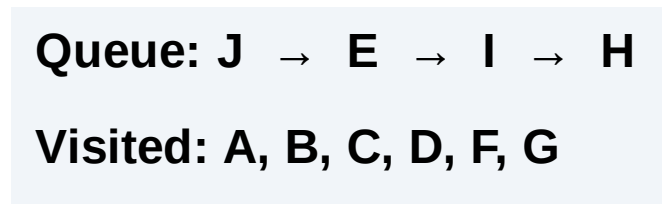
Queue: J → E → I → H

Visited: A, B, D, F, G



Step 13: No new neighbors from C

Step 13: No new neighbors from C



BFS Traversal

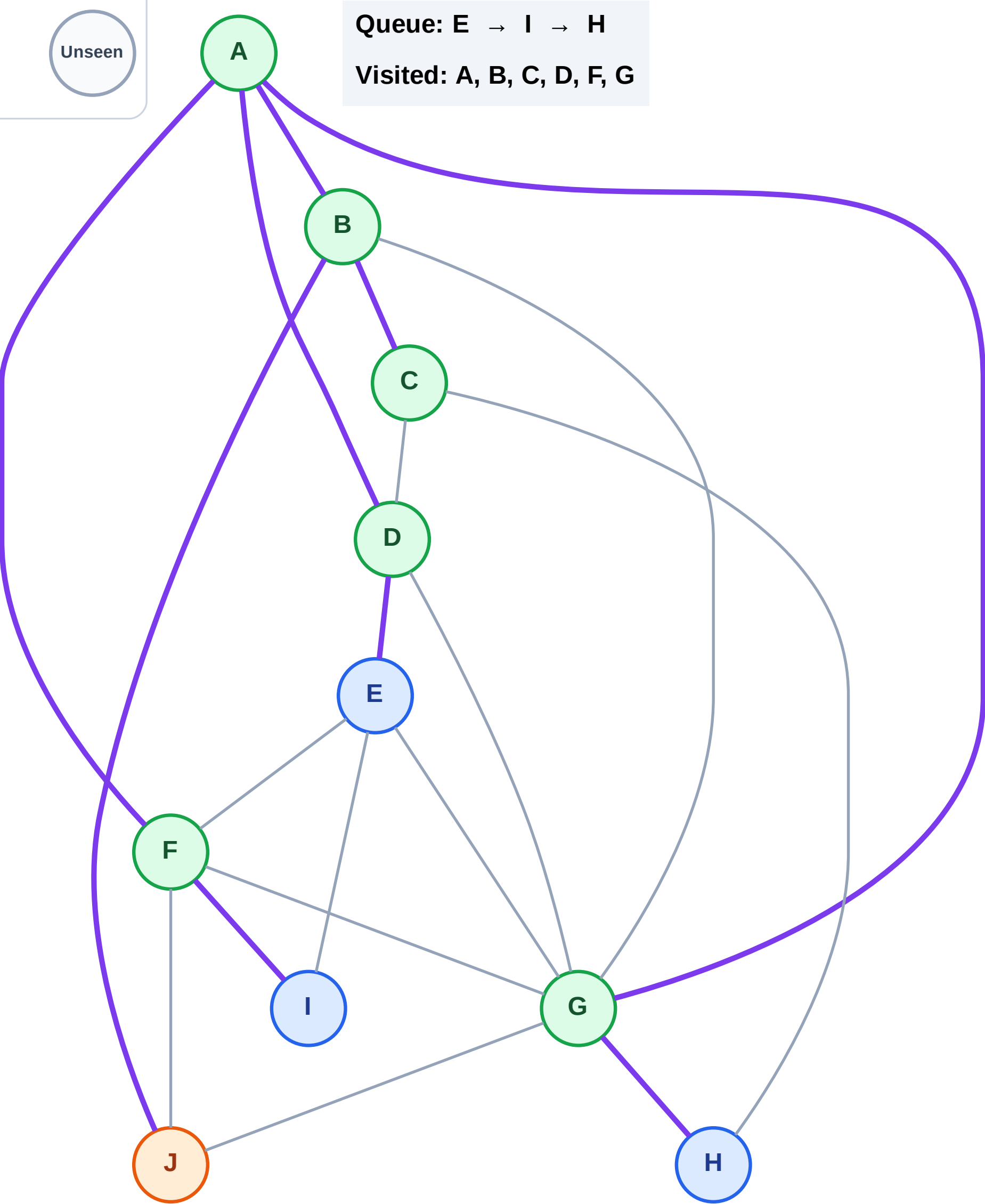
Step 14: Process node J

Legend



Queue: E → I → H

Visited: A, B, C, D, F, G



BFS Traversal

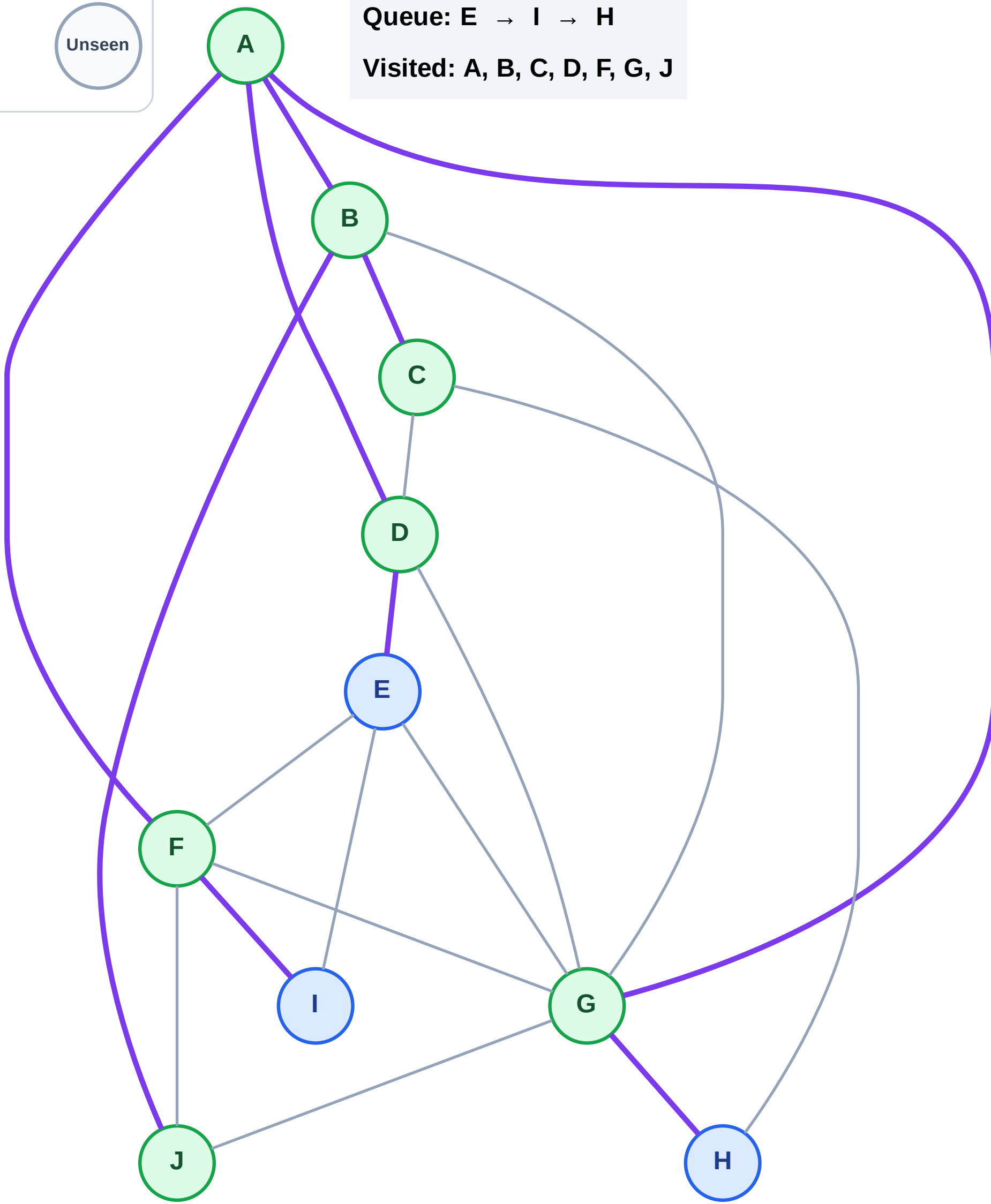
Step 15: No new neighbors from J

Legend



Queue: E → I → H

Visited: A, B, C, D, F, G, J



BFS Traversal

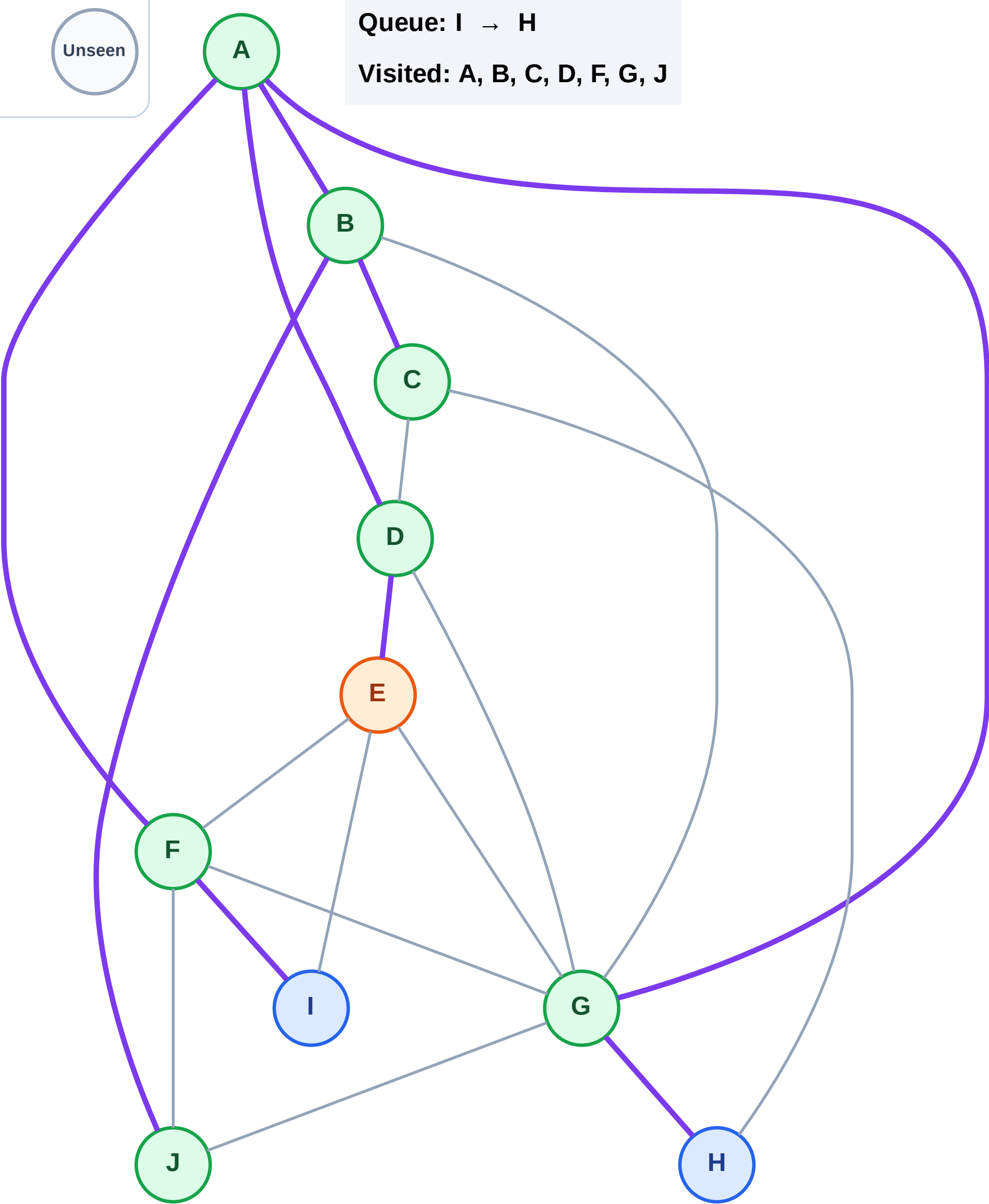
Step 16: Process node E

Legend



Queue: I → H

Visited: A, B, C, D, F, G, J



BFS Traversal

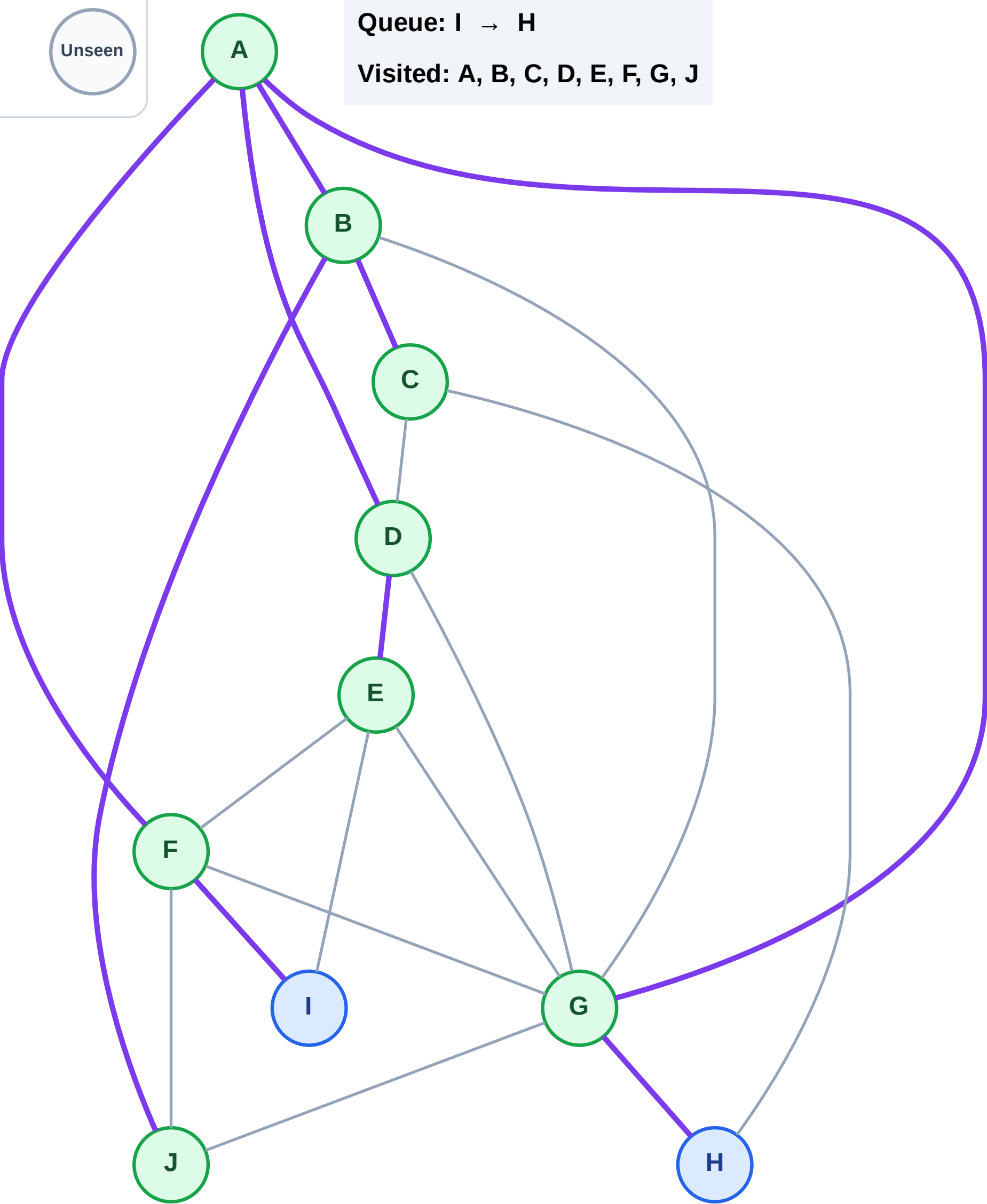
Step 17: No new neighbors from E

Legend



Queue: I → H

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node I

Legend

Complete

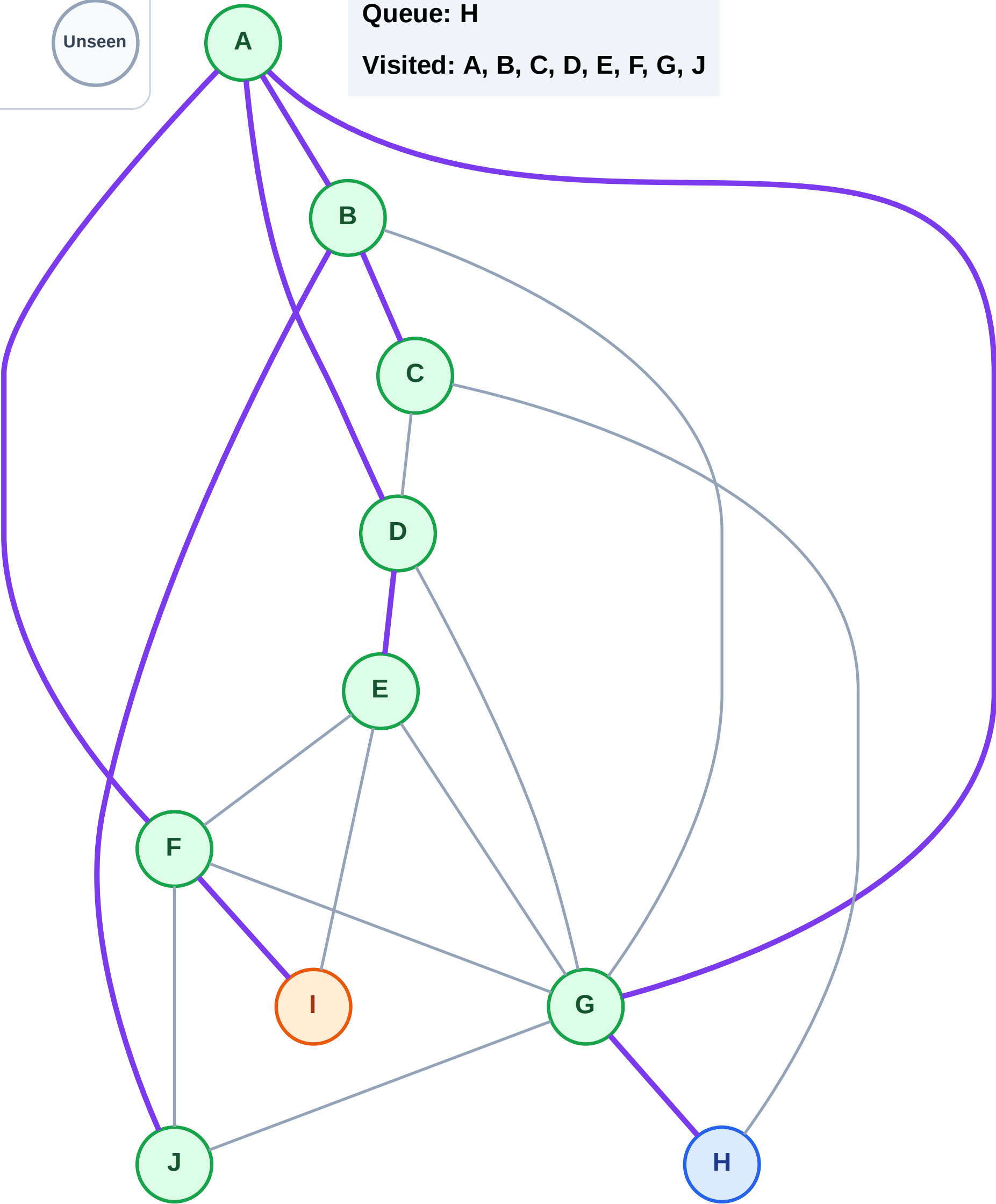
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, J



Step 19: No new neighbors from I

Step 19: No new neighbors from I



BFS Traversal

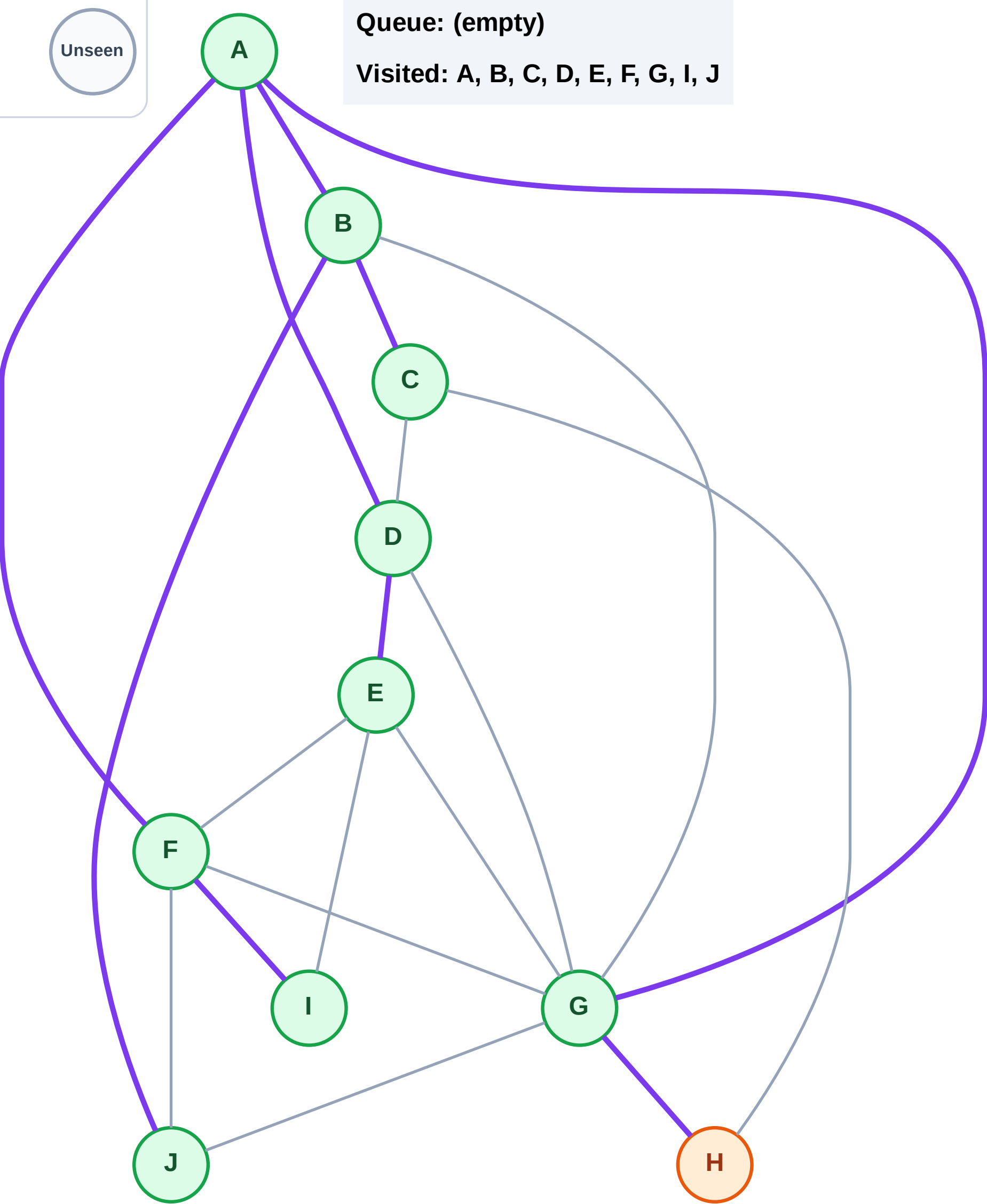
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

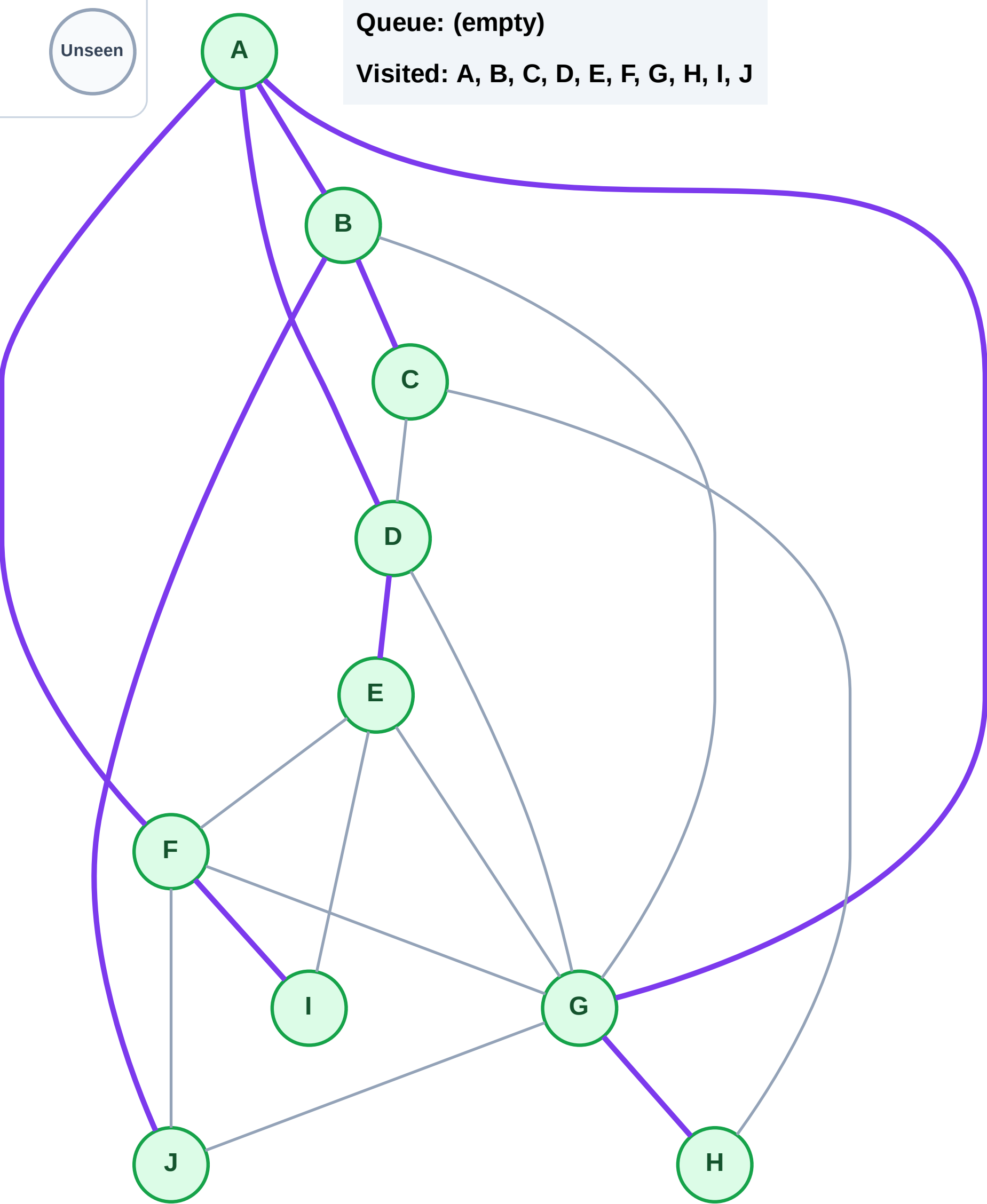
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

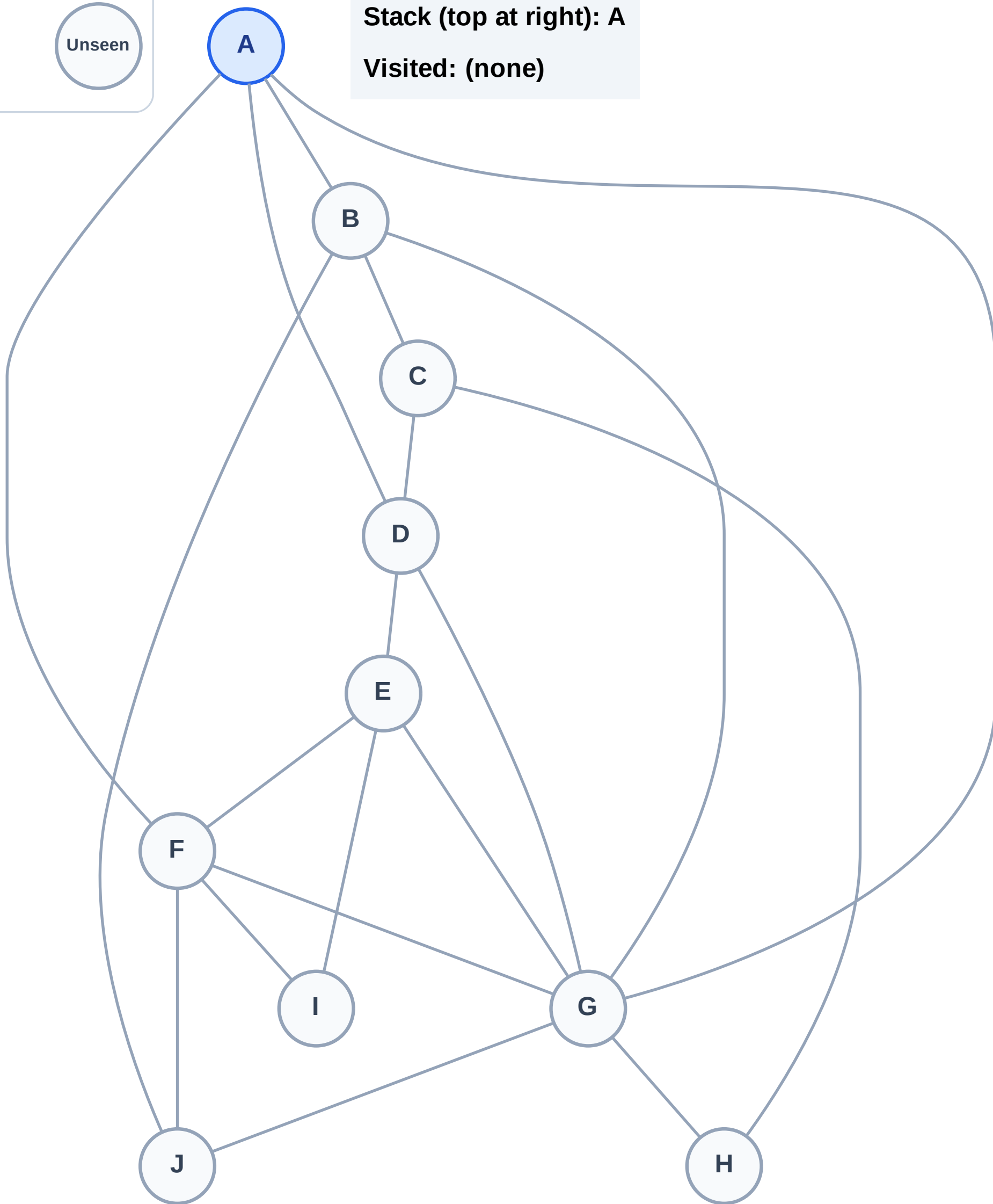
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

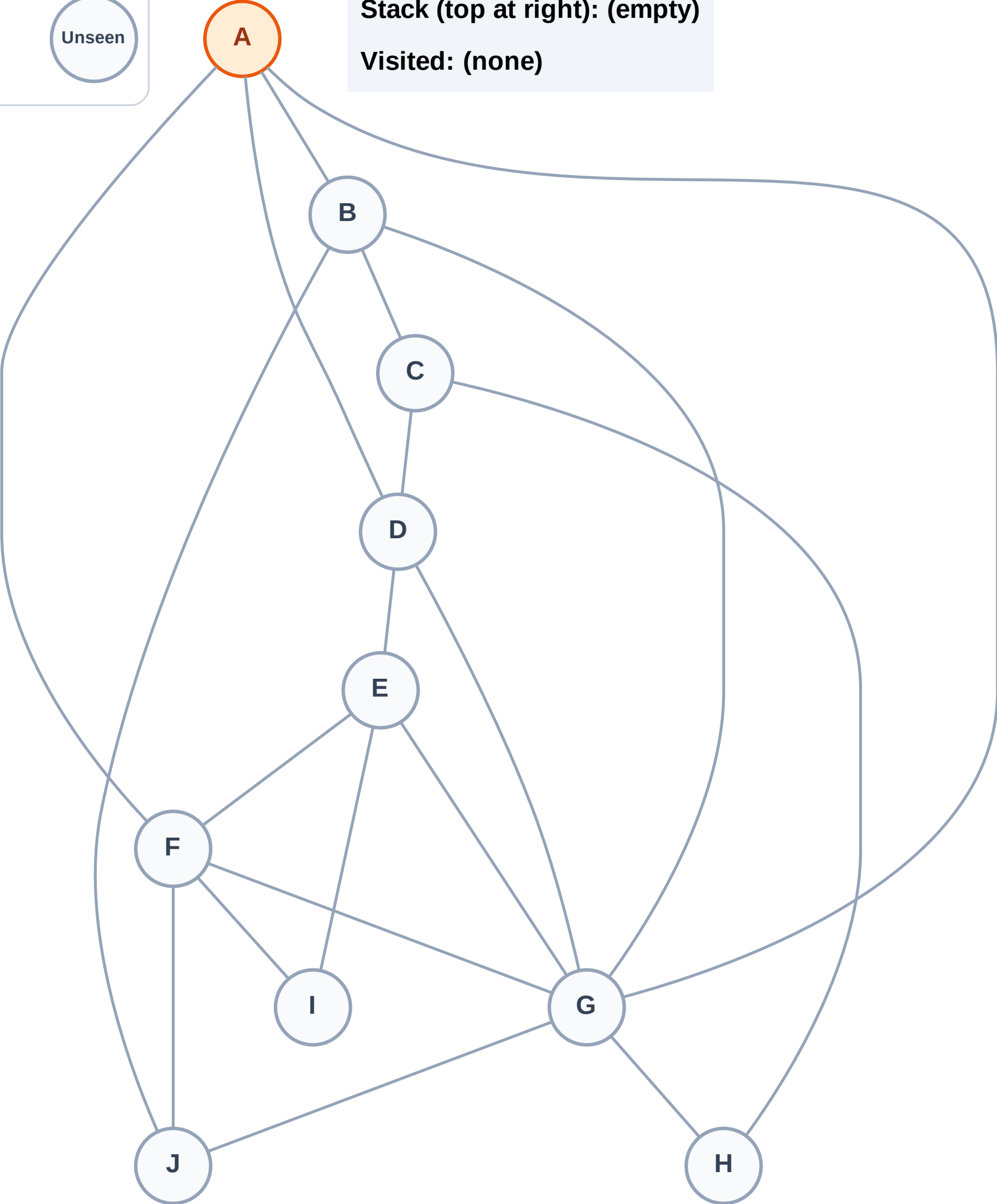
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

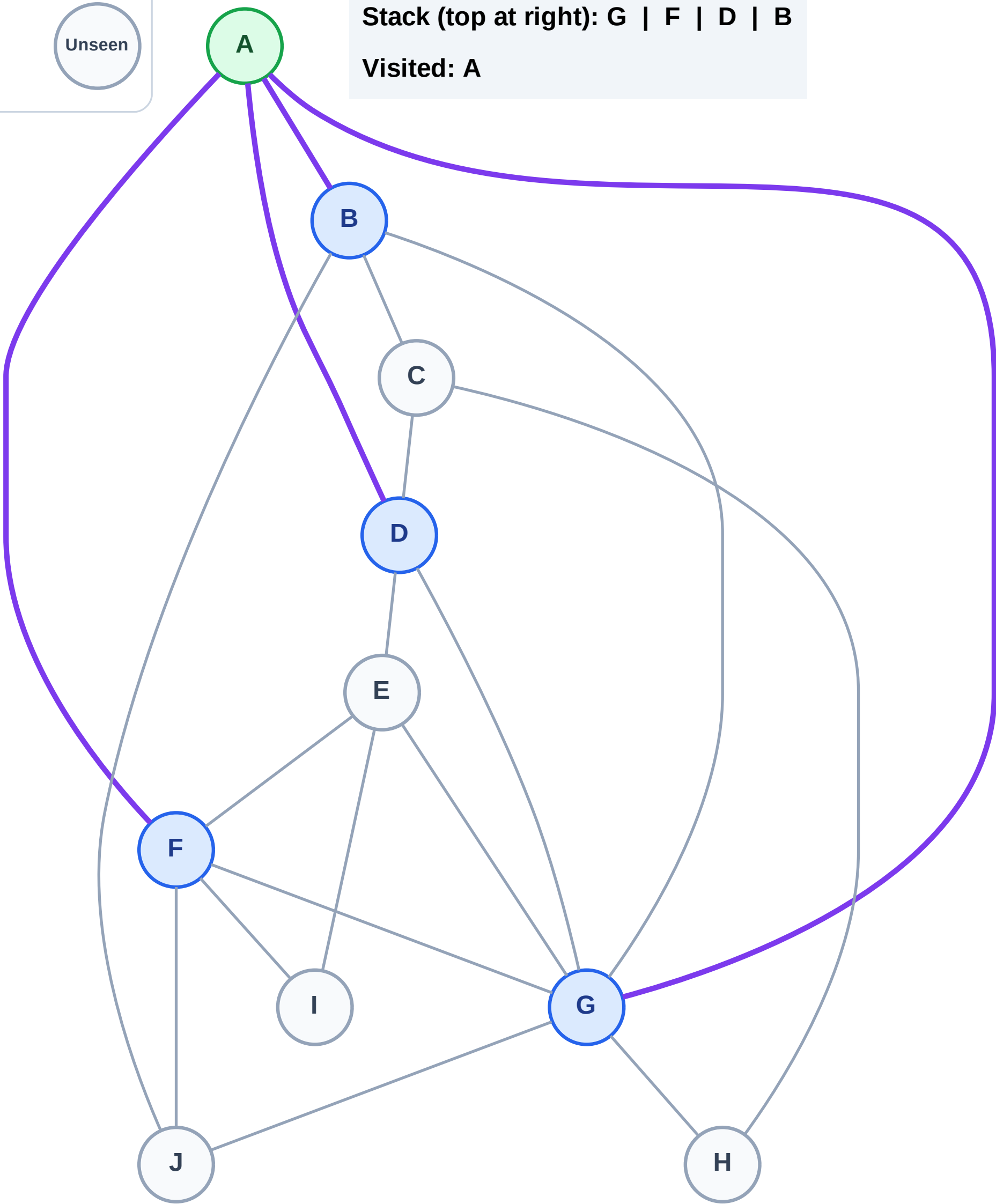
Step 3: Discover G, F, D, B from A

Legend



Stack (top at right): G | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

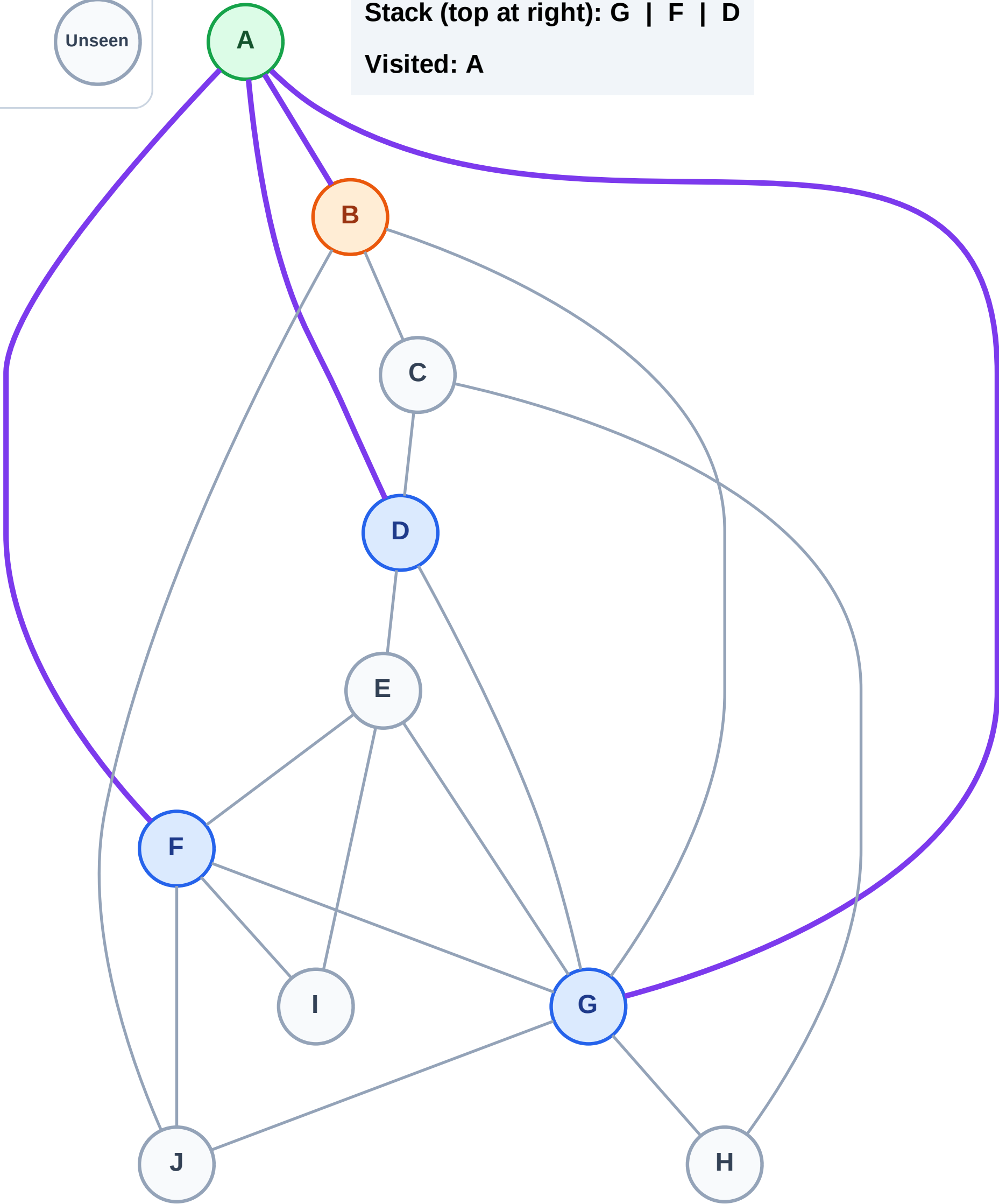
Processing

Discovered

Unseen

Stack (top at right): G | F | D

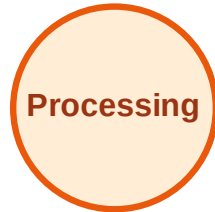
Visited: A



DFS Traversal

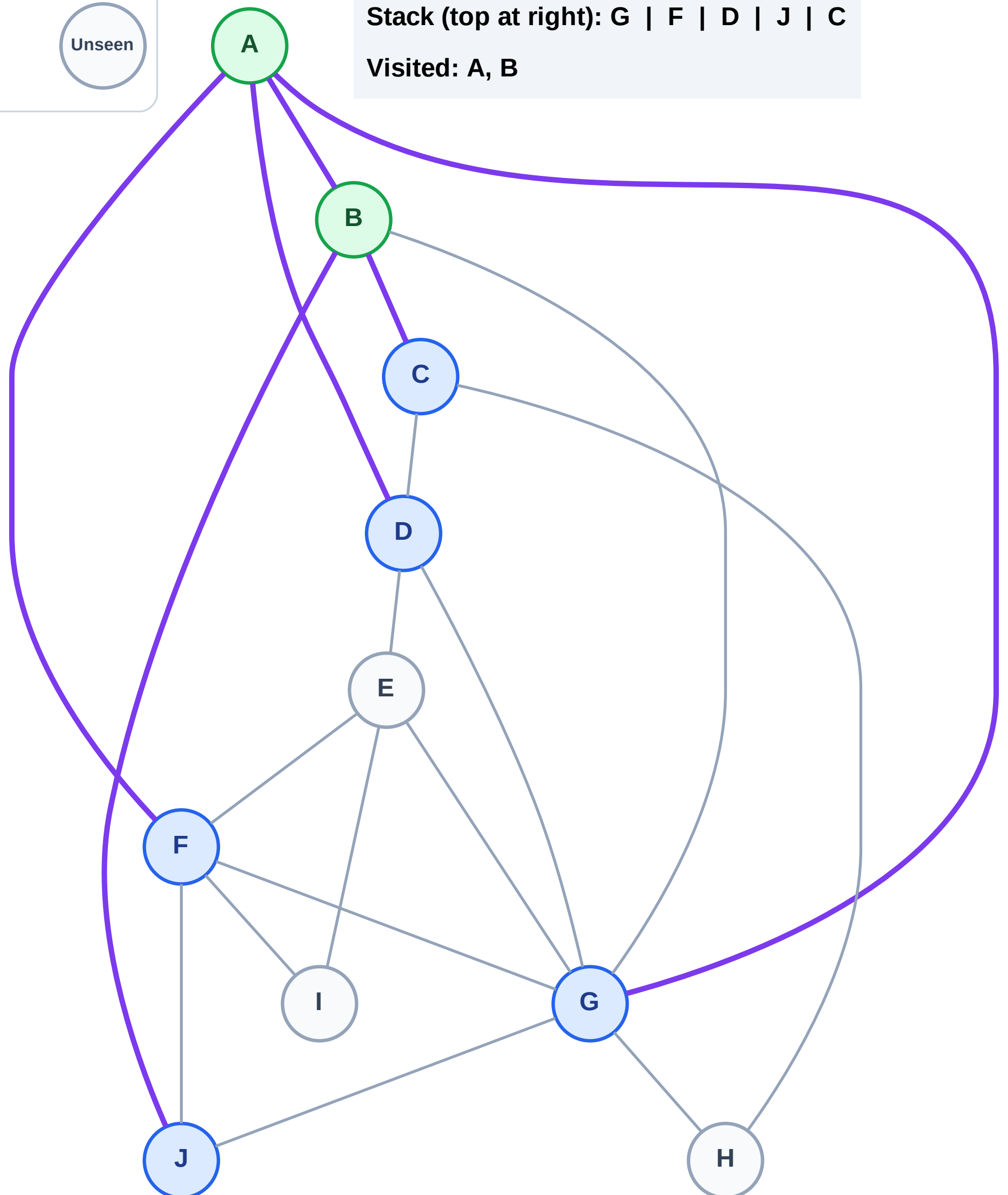
Step 5: Discover J, C from B

Legend



Stack (top at right): G | F | D | J | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

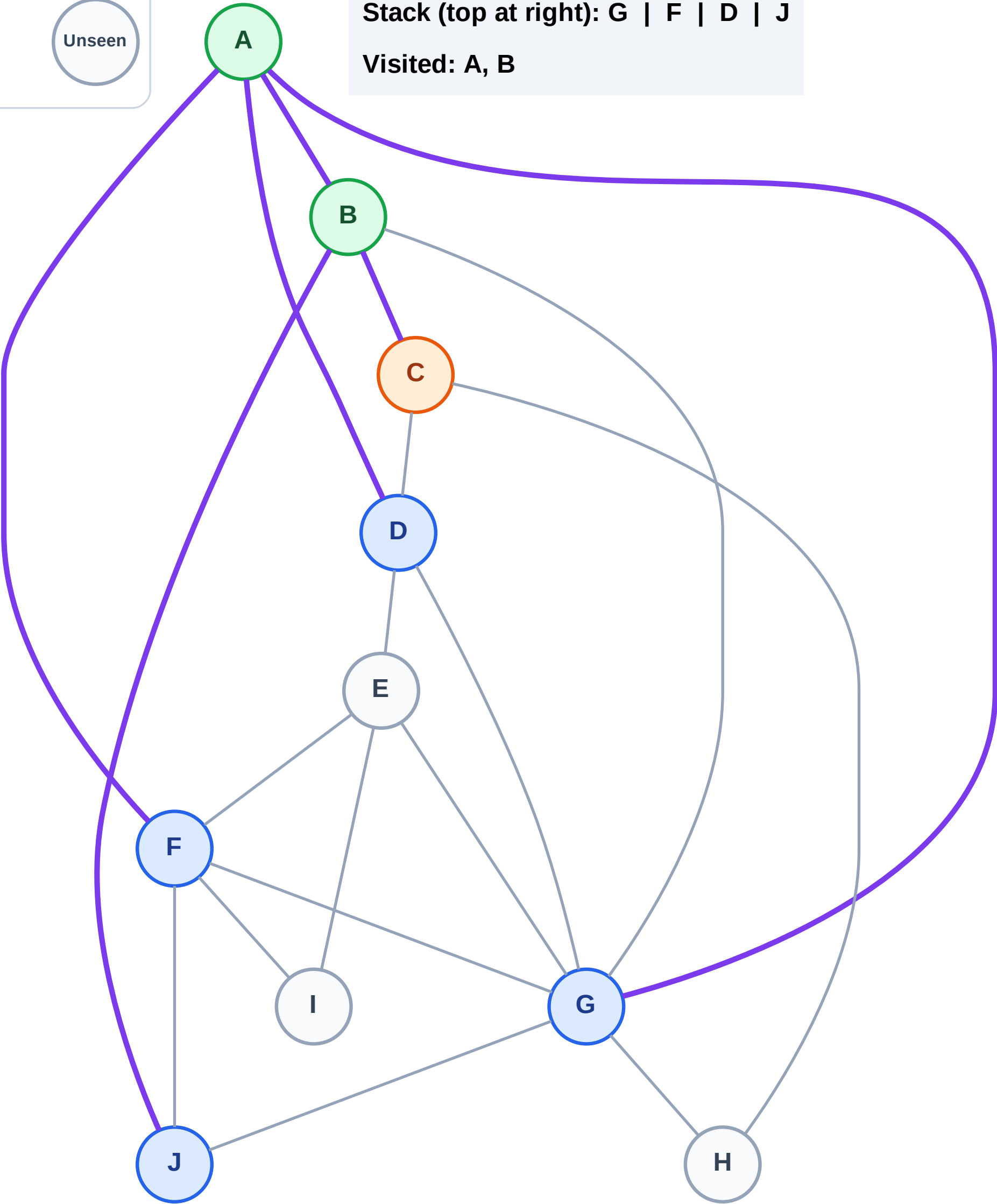
Processing

Discovered

Unseen

Stack (top at right): G | F | D | J

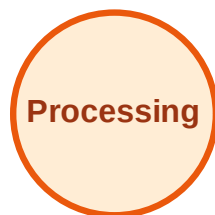
Visited: A, B



DFS Traversal

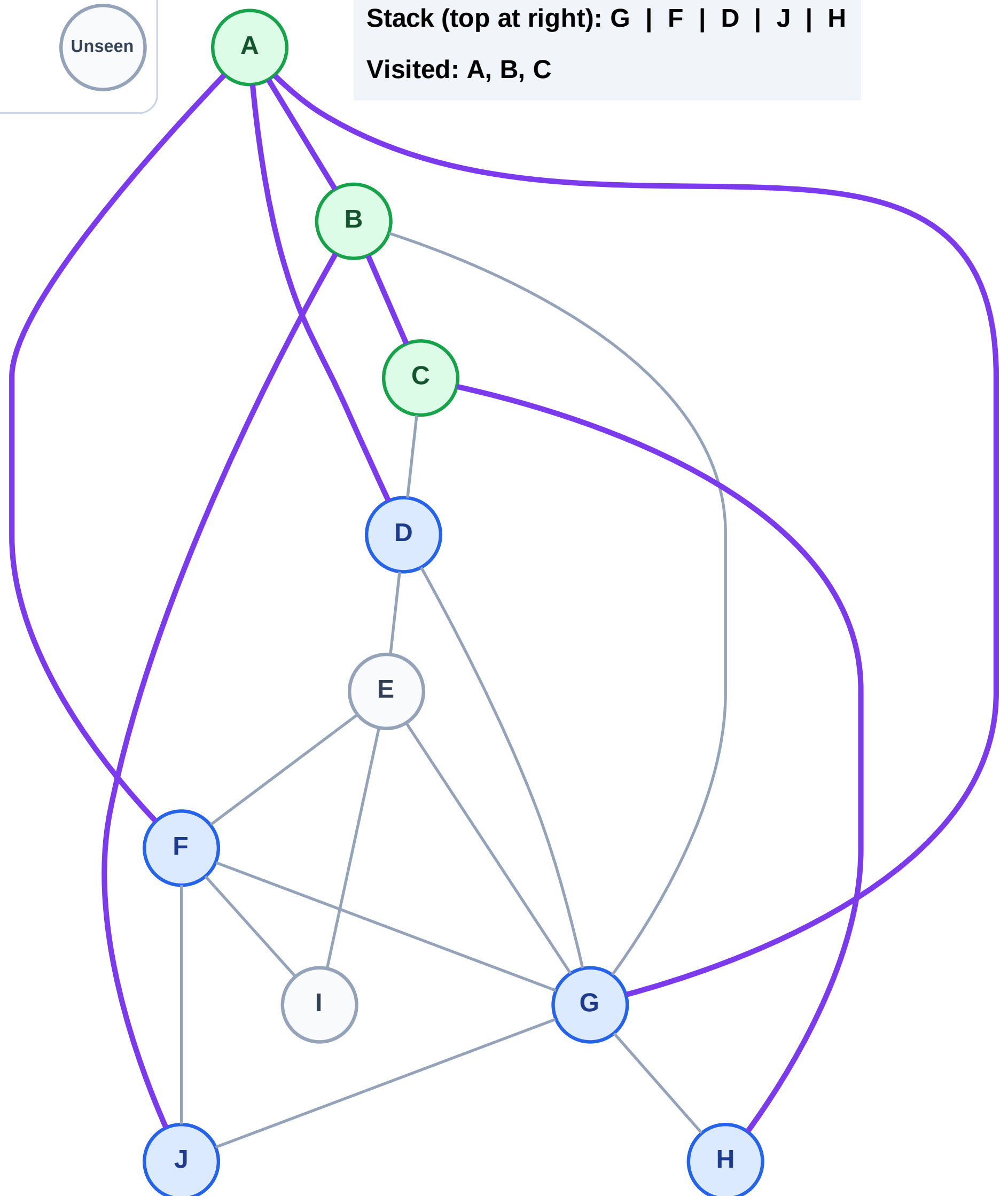
Step 7: Discover H from C

Legend



Stack (top at right): G | F | D | J | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

Complete

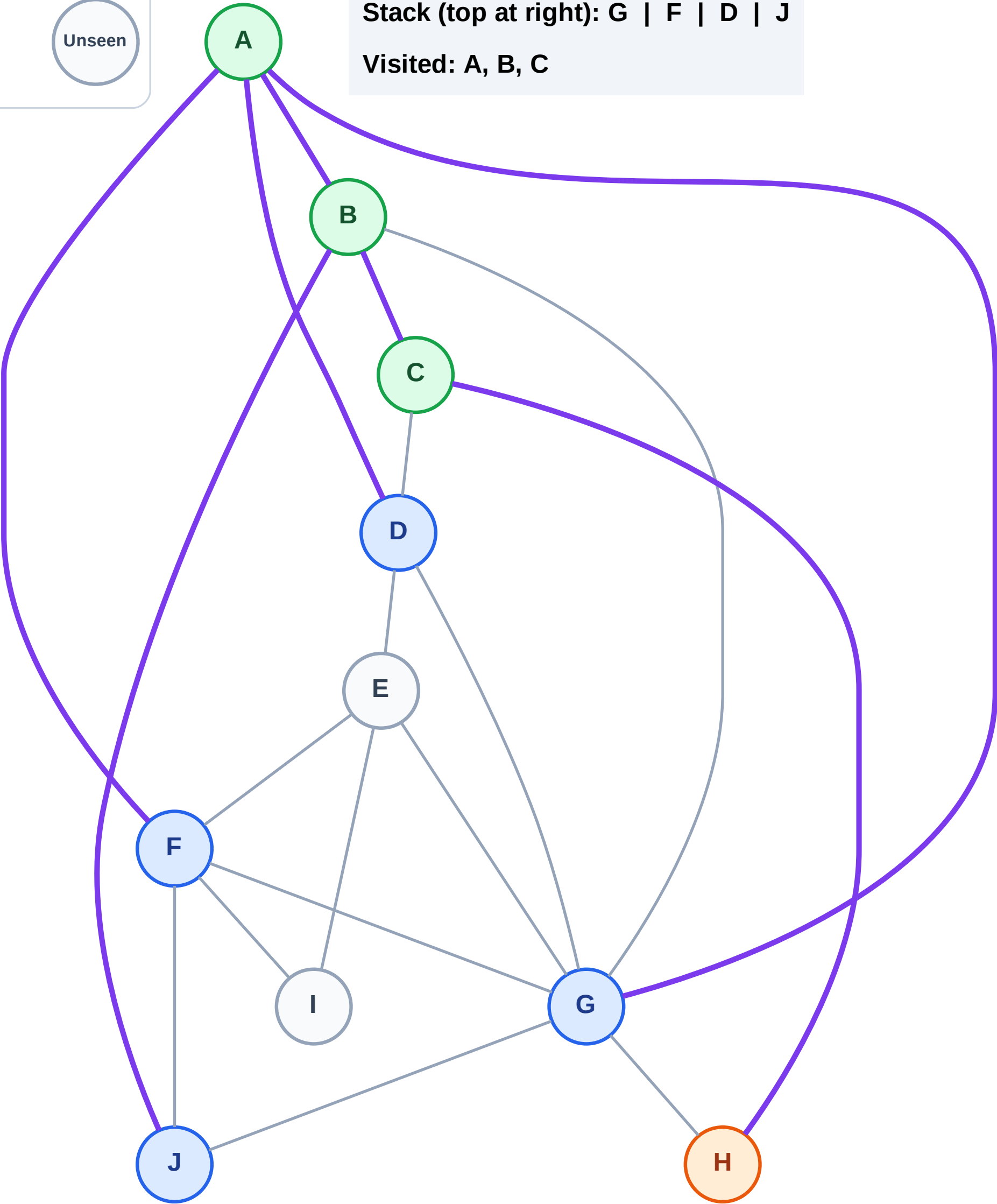
Processing

Discovered

Unseen

Stack (top at right): G | F | D | J

Visited: A, B, C



DFS Traversal

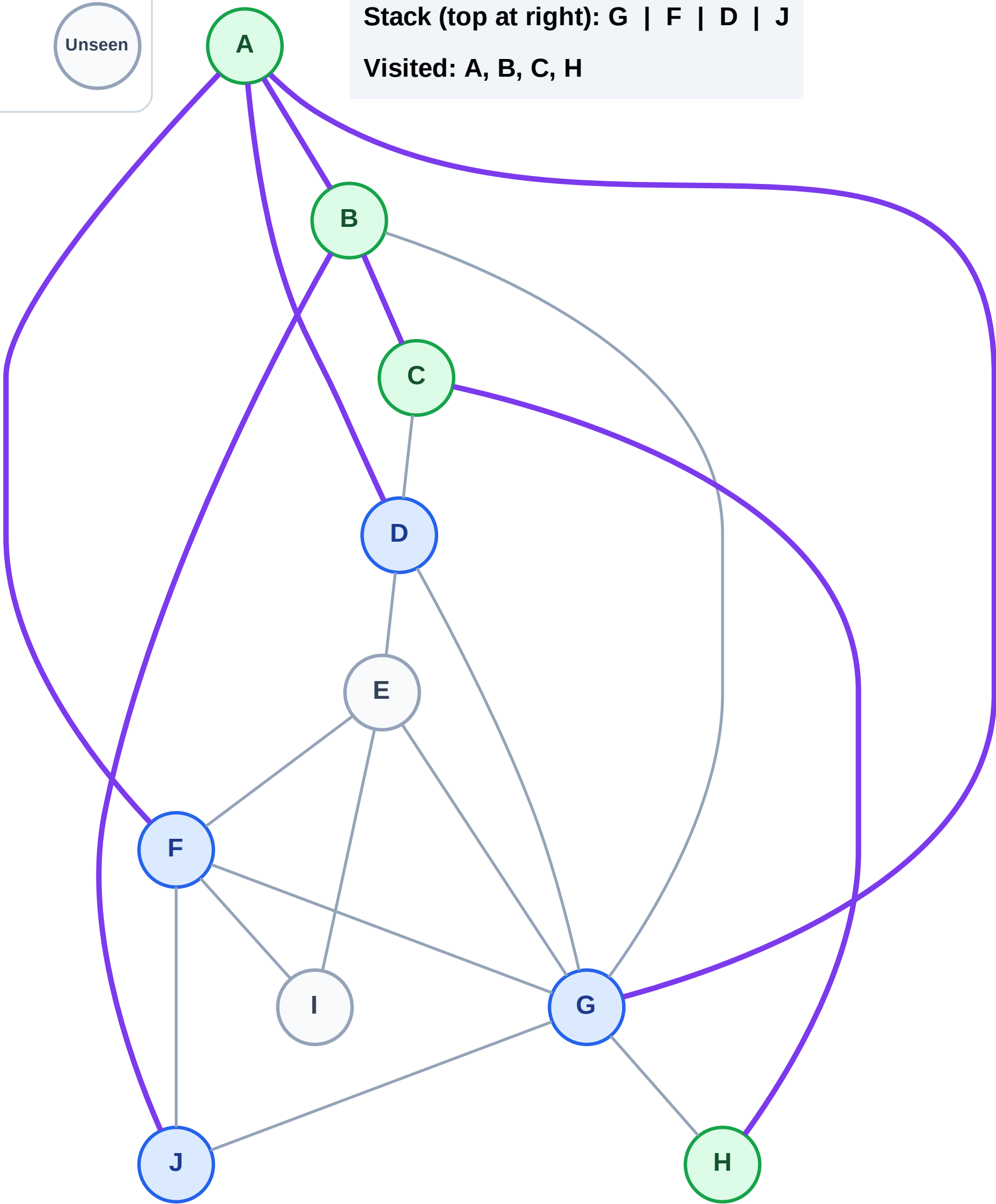
Step 9: No new neighbors from H

Legend



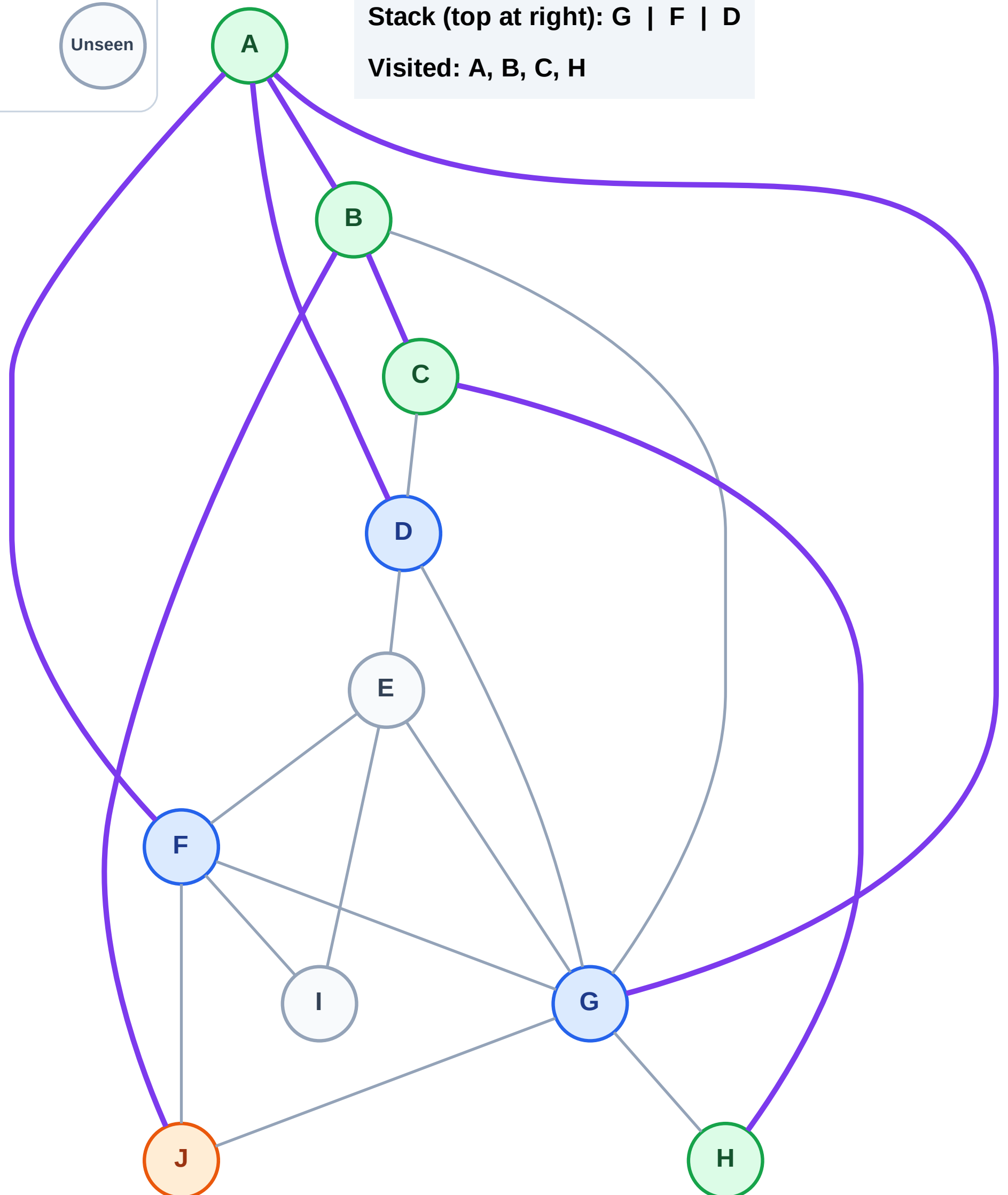
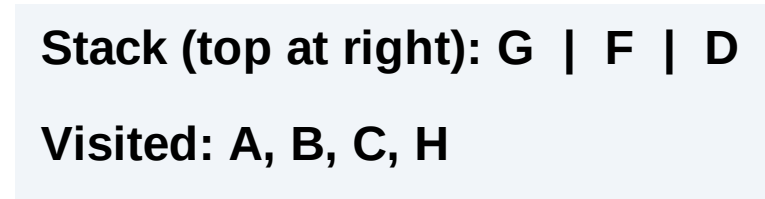
Stack (top at right): G | F | D | J

Visited: A, B, C, H



Step 10: Process node J

Step 10: Process node J



DFS Traversal

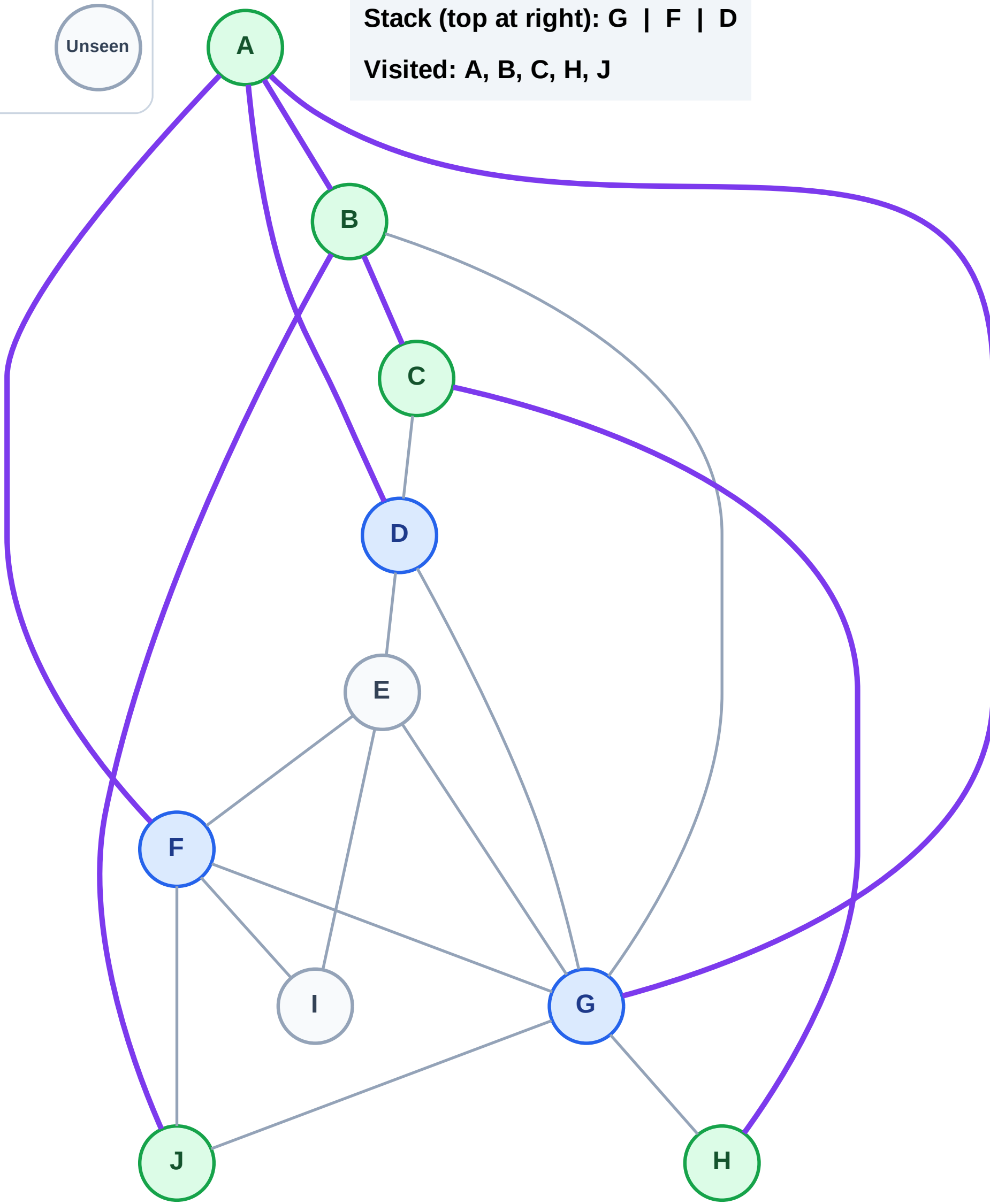
Step 11: No new neighbors from J

Legend



Stack (top at right): G | F | D

Visited: A, B, C, H, J



DFS Traversal

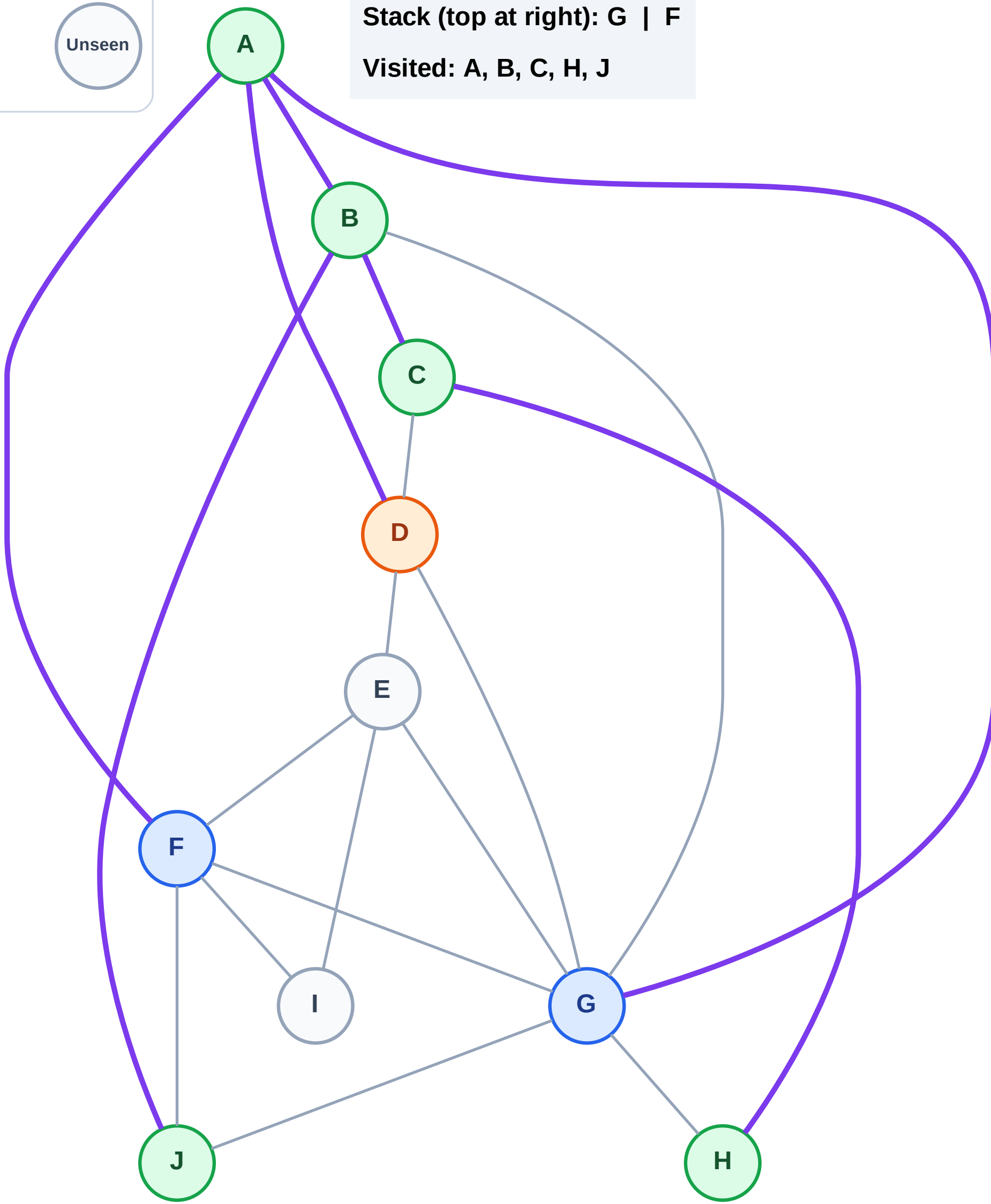
Step 12: Process node D

Legend



Stack (top at right): G | F

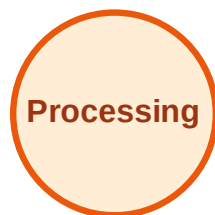
Visited: A, B, C, H, J



DFS Traversal

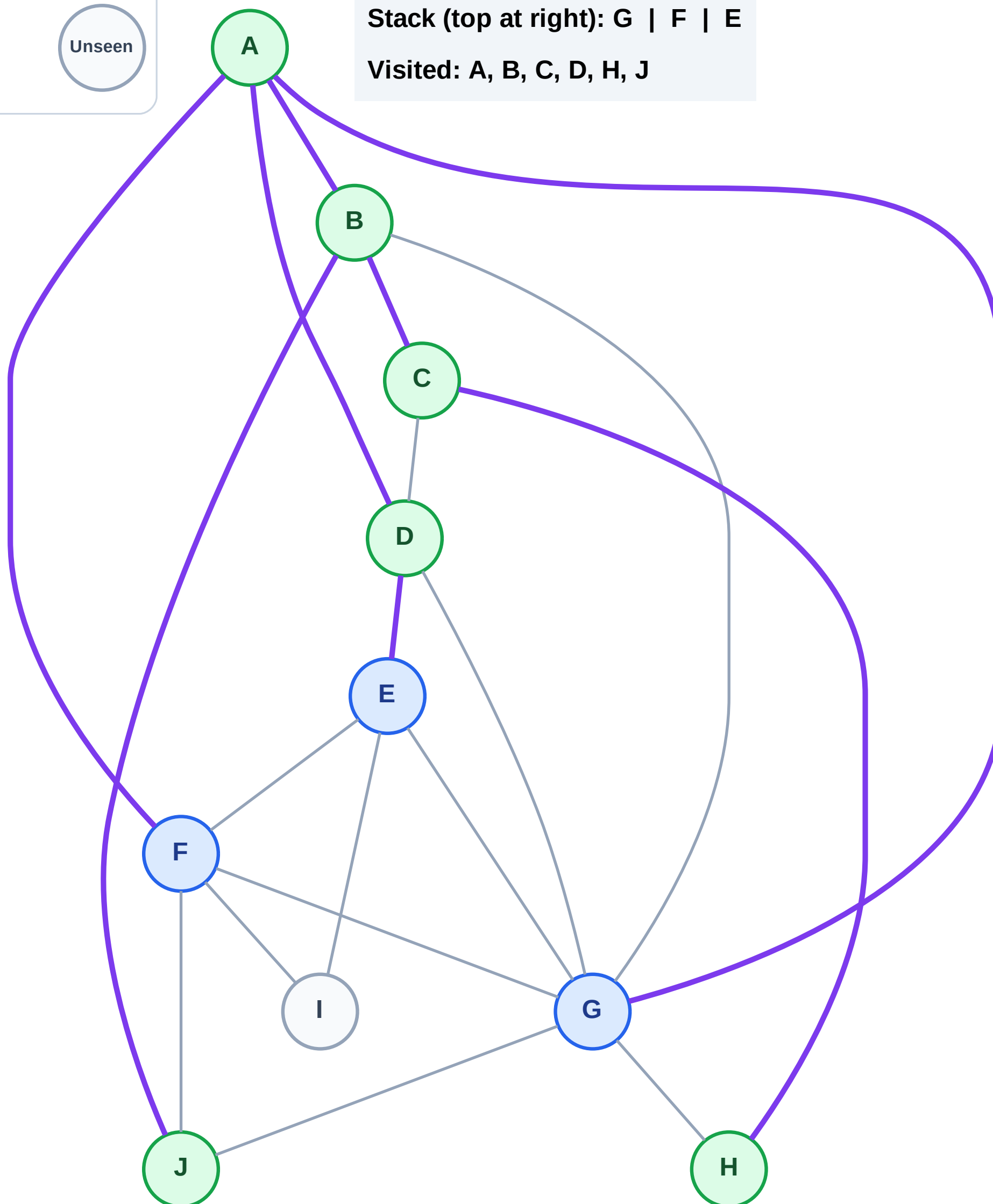
Step 13: Discover E from D

Legend



Stack (top at right): G | F | E

Visited: A, B, C, D, H, J



DFS Traversal

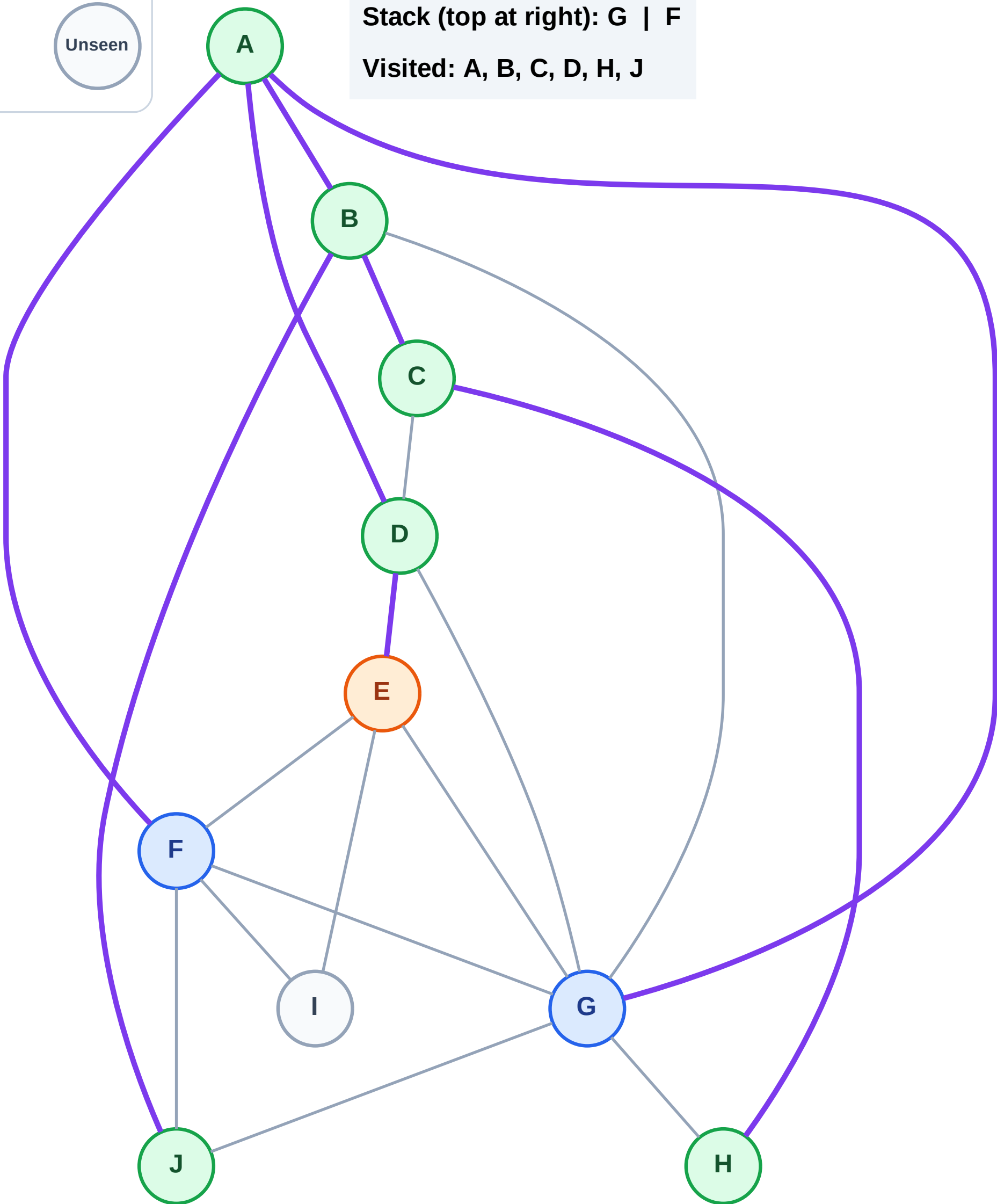
Step 14: Process node E

Legend



Stack (top at right): G | F

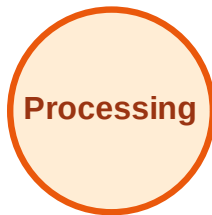
Visited: A, B, C, D, H, J



DFS Traversal

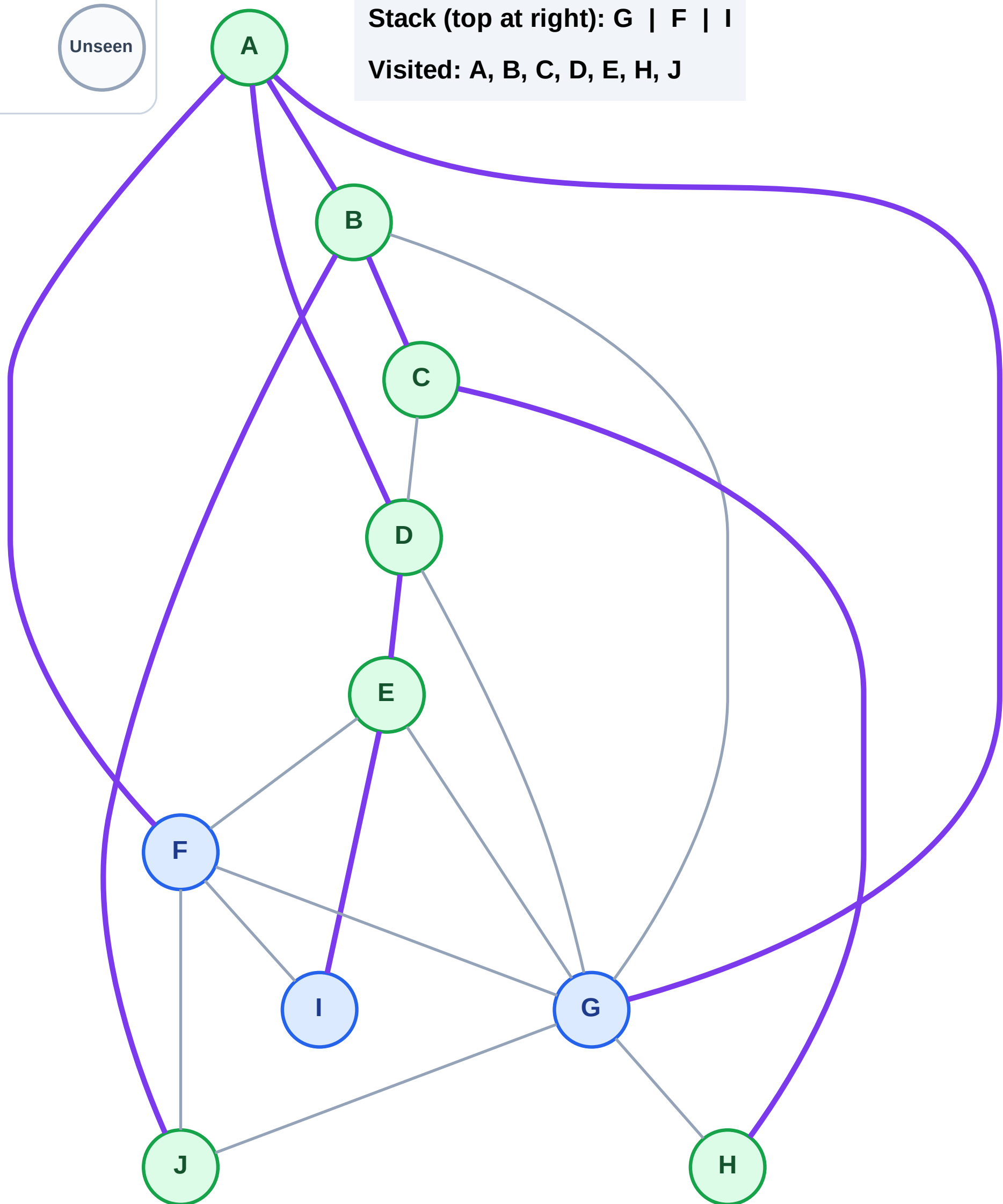
Step 15: Discover I from E

Legend



Stack (top at right): G | F | I

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node I

Legend

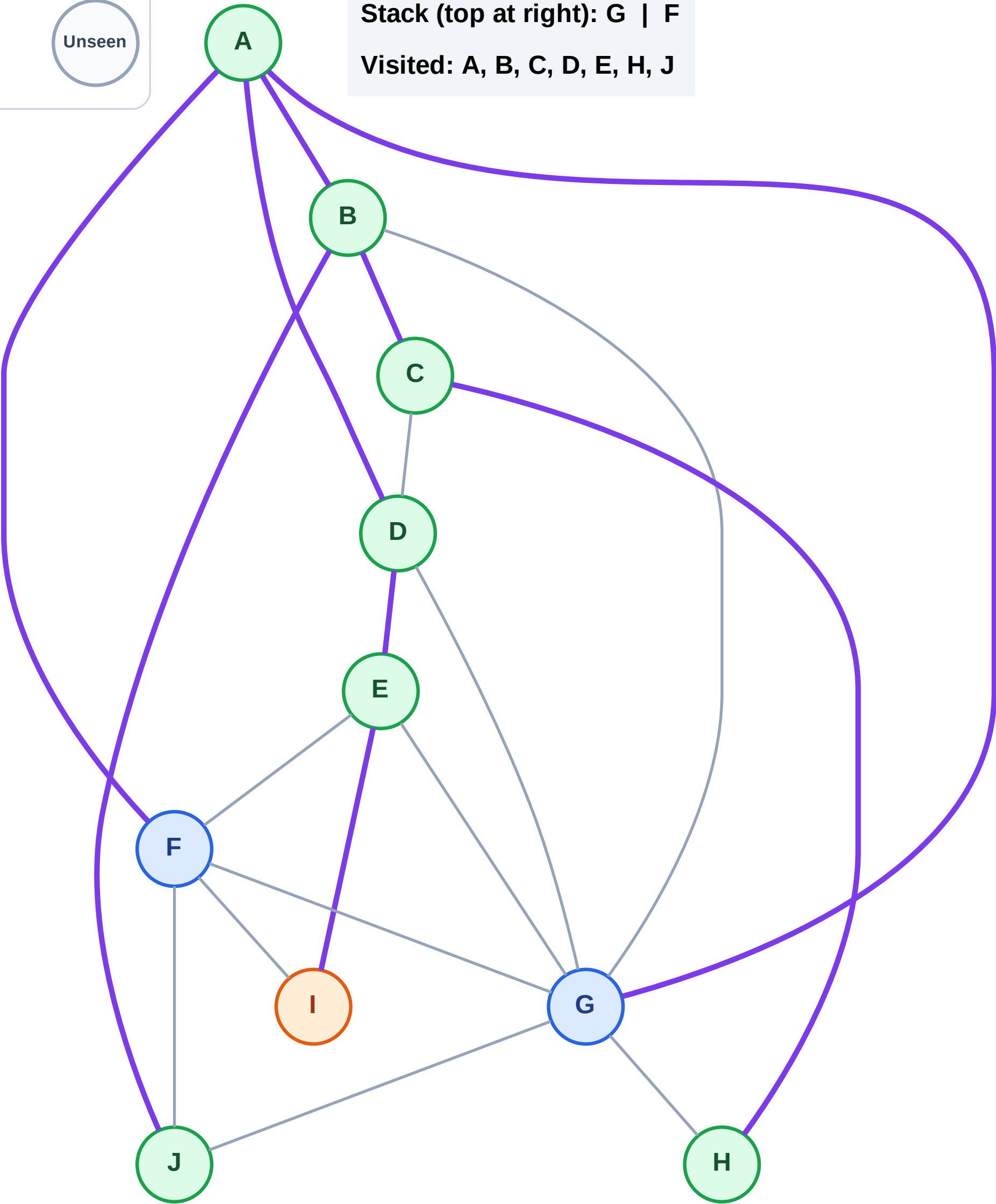
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F
Visited: A, B, C, D, E, H, J



DFS Traversal

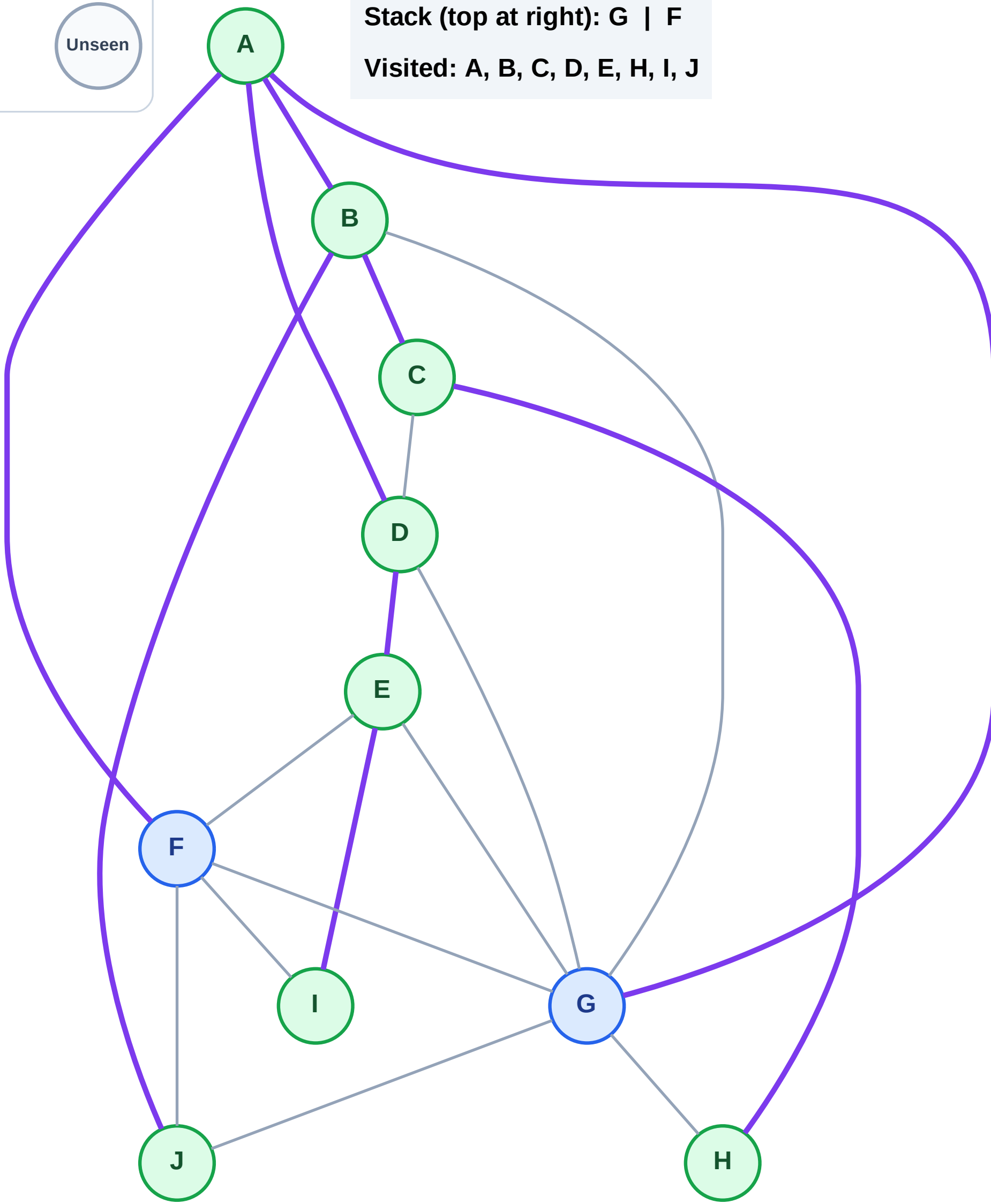
Step 17: No new neighbors from I

Legend



Stack (top at right): G | F

Visited: A, B, C, D, E, H, I, J



DFS Traversal

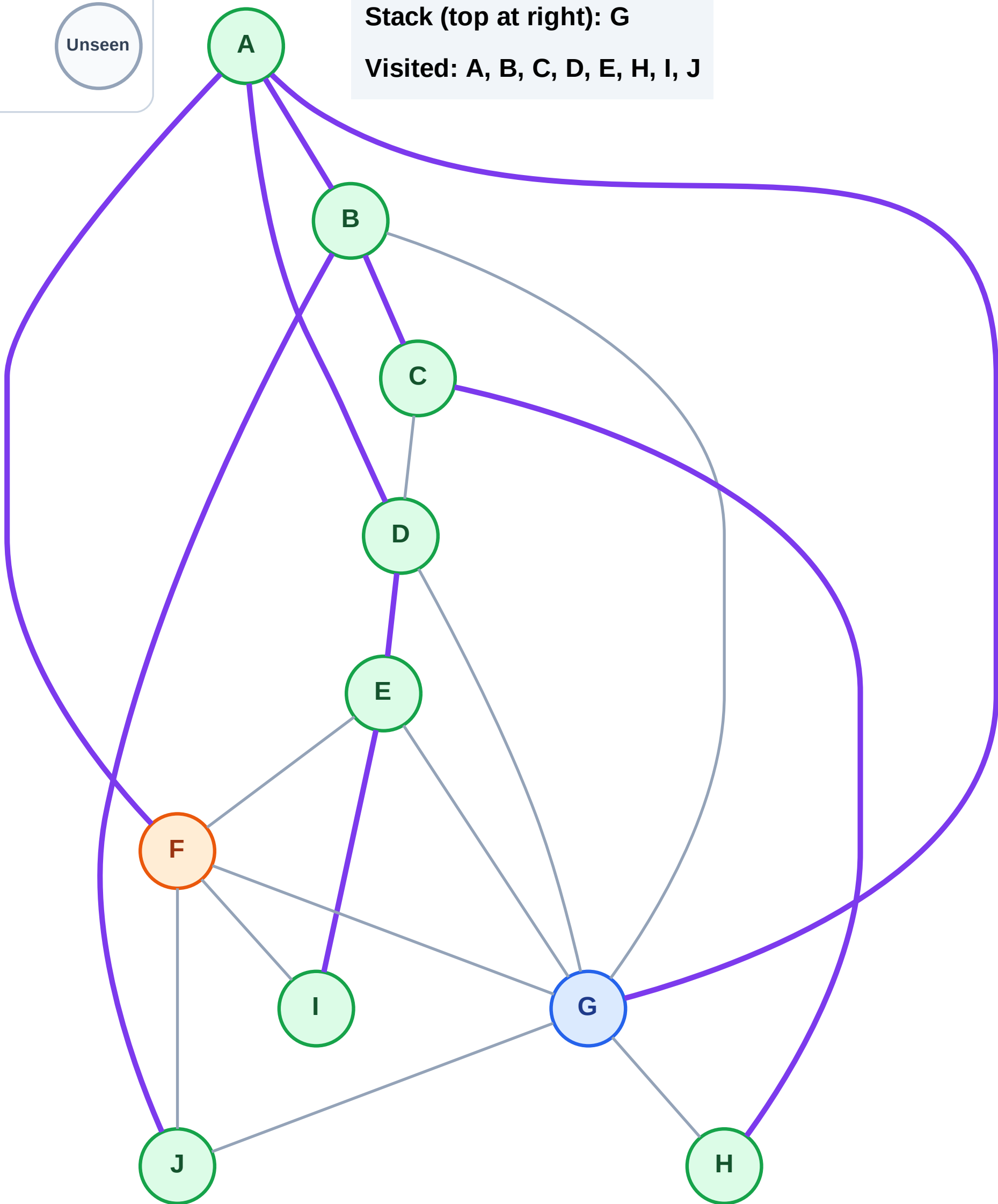
Step 18: Process node F

Legend



Stack (top at right): G

Visited: A, B, C, D, E, H, I, J



DFS Traversal

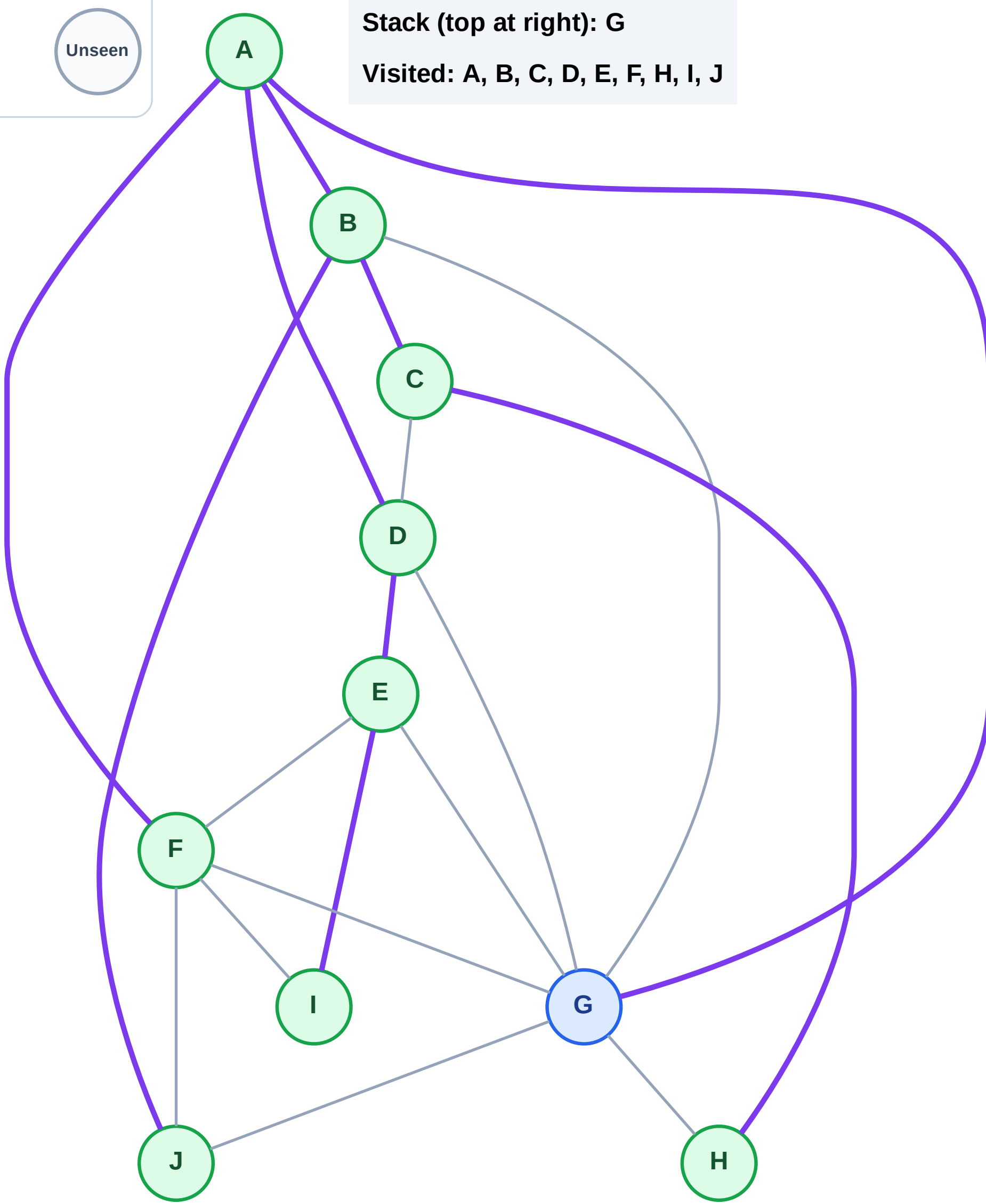
Step 19: No new neighbors from F

Legend

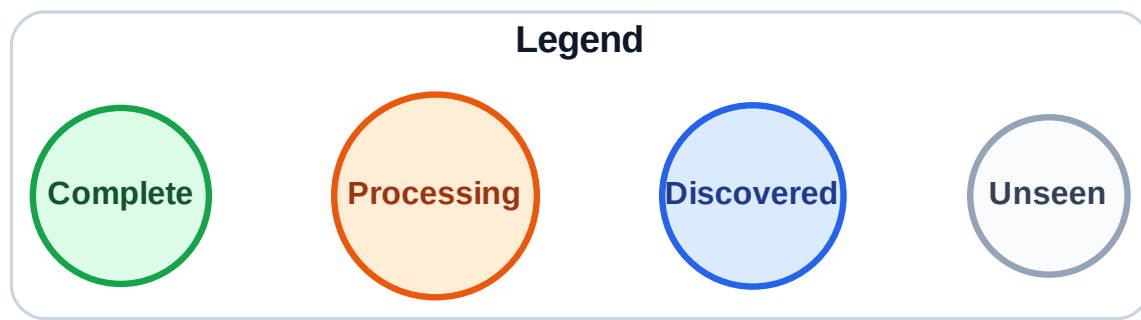


Stack (top at right): G

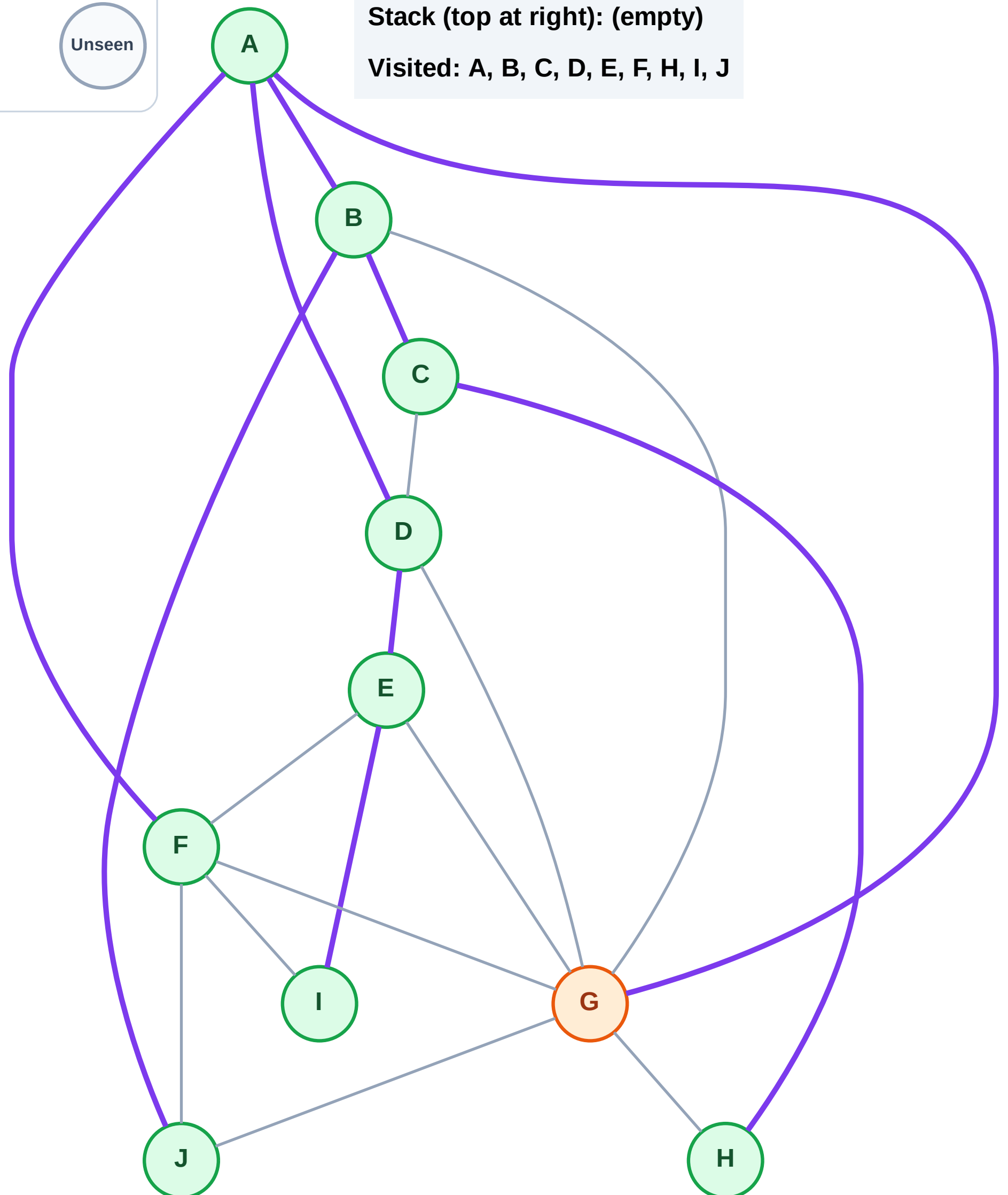
Visited: A, B, C, D, E, F, H, I, J



Step 20: Process node G



Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

